

THE BEST DISSERTATION YOU WILL EVER READ
APACHE POINT OBSERVATORY FOLLOW-UP OF
ACCELERATING CANDIDATE EXOPLANET HOST STARS (APO
ACCELS)
AND CHARACTERIZATION OF THE BINARY HOST OF THE DIRECTLY
IMAGED PLANET HD 143811 AB b

by

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Peck, A., Roberson, W., Nielsen, E., De Rosa, R., Jones, N., Wang, J., Macintosh, B., Lewis, B., Duchêne, G., Metchev, S., Abbas, A., Xuan, J., Sanghi, A., Patience, J., Barman, T., Bulger, J., Chilcote, J., Esposito, T., Fitzgerald, M., Follette, K., Gallamore, H., Goodsell, S., Graham, J., Greenbaum, A., Hiben, P., Ingraham, P., Kalas, P., Konopacky, Q., Marchis, F., Maire, J., Marois, C., Matthews, B., Mawet, D., Millar-Blanchaer, M., Oppenheimer, R., Palmer, D., Perrin, M., Poyneer, L., Pueyo, L., Rajan, A., Rameau, J., Rantakyö, F., Ren, B., Ruffio, J.B., Savransky, D., Schneider, A., Sivaramakrishnan, A., Smith, A., Song, I., Soummer, R., Thomas, A., Ward-Duong, K., Wolff, S., *Characterization of the Host Binary of the Directly Imaged Exoplanet HD 143811 AB b*. ApJ Letters, 995, 40, doi: 10.3847/2041-8213/ae1a67

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CONFERENCE PROCEEDINGS

AWARDS AND GRANTS

2021–2023	NMSU Astronomy William Webber Voyager Graduate Fellowship
2025–2026	NMSU Astronomy Pegasus Award for excellence in teaching
2025-2026	NMSU Preparing Future Faculty Fellowship

LEADERSHIP AND PROFESSIONAL EXPERIENCE

ABSTRACT

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Dr. Eric Nielsen, Advisor

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LIST OF ABBREVIATIONS

Instruments, Telescopes & Observatories

Fermi	Fermi Gamma-ray Space Telescope
INTEGRAL	International Gamma-ray Astrophysics Laboratory
KAGRA	Kamioka Gravitational Wave Detector
LIGO	Laser Interferrometer Gravitational Wave Observatory
LISA	Laser Interferometer Space Antenna
LVK	LIGO-Virgo-KAGRA
Virgo	Virgo interferometer
ZTF	Zwicky Transient Facility

Acronyms and Jargon

1g, 2g, 3g	first, second, third generation black hole
AGN	active galactic nucleus
BBH	binary black hole
bf	bound-free
BH	black hole
BLR	broad line region
CO	compact object
EM	electromagnetic
EMRI	extreme-mass ratio inspriial
ff	free-free
GRB	gamma-ray burst
GW	gravitational wave
HLLE	Harten-Lax-van Leer-Einfeldt solver
IMBH	intermediate mass black hole
IMF	initial mass function
IMRI	intermediate-mass ratio inspiral
ISCO	inner-most stable circular orbit
LSST	Legacy Survey of Space and Time
McFACTS	Monte Carlo for AGN Channel Testing and Simulations
NSC	nuclear star cluster
PDS	pressure-driven snowplow
PISN	pair-instability supernova
O3, O4, O5	LVK observing runs 3, 4, and 5
SG	Sirko and Goodman disk model
SMBH	supermassive black hole
SLSN	superluminous supernova
SN	supernova
TQM	Thompson, Quataert, and Murray disk model

LIST OF SYMBOLS

Symbol	Meaning
$q = M_2/M_1, \ M_1 > M_2$	mass ratio
M_1	mass of more massive binary component
M_2	mass of less massive binary component

1. TITLE

2. Introduction

imaging of exoplanets allows for demographic studies of wide-separation giant planets and for analysis of these planets' atmospheres through spectra and photometry (e.g. ??). Each direct imaging detection allows for detailed characterization of the companion (e.g. ?????). While wide-separation giant planets have relatively low occurrence rates (e.g. ??), additional companions can be identified from astrometric accelerations of their host stars (e.g. ?). A targeted survey informed by astrometric accelerations, radial velocities, and stellar ages can complement previous blind surveys by increasing the number of directly imaged substellar companions. Planets detected via astrometry have the added benefit of measurable dynamical masses making them key benchmarks for planet formation and evolution models (e.g. ???), especially if their ages are known (e.g. ??). The mass of a companion can be determined by combining imaging and astrometry, and the companion's age can be measured from observations of the host star. A broader sample of benchmark exoplanets will help constrain planet formation and evolution models.

Gaia astrometry is expected to be sensitive to planets between $\sim 1 - 15 M_{\text{Jup}}$ with periods between $\sim 0.5 - 10$ yr around a wide range of stellar host masses (?). *Gaia* Data Release 4 (DR4) is scheduled for mid-2026 and is set to include a list of

astrometrically detected exoplanets between $\sim 1 - 5$ AU detected using only *Gaia* (?). Prior to the release of DR4, substellar companions can be detected by combining positions, parallaxes, and proper motions from the *Hipparcos* Catalog (??) and *Gaia* Data Release 3 (DR3) (?). Using astrometry the influence of a massive planet on the path of a star can be observed, and longer-period companions can be detected by linking these two missions (e.g. ???). Velocity is measured as proper motion for stars in both *Hipparcos* (1991.25) and *Gaia* DR3 (2016.0) as well as from the difference between the two catalog positions. An acceleration of the photocenter can be identified from discrepancies between the three velocity vectors. Combined *Hipparcos* and *Gaia* astrometry yield typical acceleration precision as low as $\sim 1 \text{ m s}^{-1} \text{ yr}^{-1}$ for nearby stars, and is sensitive to longer periods than *Gaia* alone (?).

Several substellar companions have been detected using this method. The $3 - 7 M_{\text{Jup}}$ planet AF Lep b was identified by the acceleration of its young host star, a member of the ~ 24 Myr β Pictoris moving group (?), before being detected by direct imaging (???). Higher mass substellar companions have been imaged around older accelerating stars including the $13.9 - 16.1 M_{\text{Jup}}$ companion to the ≤ 500 Myr HIP 99770 (?), the $\sim 28 M_{\text{Jup}}$ companion to the ~ 750 Myr HIP 21152 (?), and the $\sim 31 M_{\text{Jup}}$ companion to the $\lesssim 1.5$ Gyr HIP 5319 (?). Since these accelerations are from astrometric measurements of position and velocity with a limited time baseline, there is a degeneracy between the companion mass

and orbital parameters, especially the period. For example, the acceleration of the ~ 500 Myr HR 1645 appeared to be consistent with a brown dwarf companion on a ~ 100 yr orbit but was instead shown to be caused by a stellar binary with a < 1 yr orbit (?).

Planets cool monotonically over time, with younger, more massive planets being brighter. Given current ground-based direct imaging sensitivity, substellar companions with a contrast ratio brighter than 10^{-6} can generally be detected to within $\sim 0.5''$ of their host stars (e.g. ??). Depending on the luminosity and distance of the host star, a wide separation ($\sim 10 - 30$ AU) giant planet can be detected around stars younger than a few hundred Myr, and brown dwarfs around stars younger than ~ 1 Gyr. Thus, measuring accurate ages of host stars and identifying young systems is crucial when selecting high-priority direct imaging targets. We measure the ages of 166 northern hemisphere accelerating stars so that we can identify the youngest systems and prioritize them for direct imaging follow-up.

We use the following methods to derive the ages of our sample stars.

1. **Ca H and K emission (R'_{HK}):** The calcium H and K emission lines are observed as cores within deep absorption lines. The absorption lines form in the optically thick photosphere and the emission cores are generated in the optically thin chromosphere via magnetic heating. Thus, these emission

cores trace magnetic activity in stars. As a star spins down with age, its magnetic and chromospheric activity decreases (e.g. ?). The youngest stars have the fastest rotation and, thus, the strongest magnetic activity, which causes the strongest calcium emission. R'_{HK} is calibrated as an age tracer for F5-K2 stars (e.g. ???).

2. **Lithium:** Throughout the main sequence lifetime of solar-type and lower mass stars, convection transports surface lithium to deeper, hotter layers of the star where it is destroyed via fusion. Lithium depletion is most efficient in low-mass stars that are either fully convective or have thick convective envelopes (???) making it a useful age tracer for G, K, and M stars. Lithium equivalent width is a powerful age tracer for very young late-type (later than $\sim$ K5) stars due to their rapid lithium depletion, but after about 150 Myr only equivalent width upper limits can be established for these stars.
3. **Gyrochronology:** Stars later than $\sim$ F7 spin down as they age due to angular momentum loss via magnetic braking (?). Rotation periods can be measured from flux variations due to star spots rotating on and off the disk of the star. Longer rotation periods correspond to older ages.
4. **Color-Magnitude Diagram Positions:** After reaching the zero-age main sequence, the position of a star on the CMD does not change significantly during the first third of its main sequence lifetime. During the final two-

thirds of its main sequence lifetime, the core becomes more helium-rich, increasing the star’s luminosity. This results in detectable movement across the CMD. Age posteriors for these stars can be derived by comparing their photometry to theoretical isochrones (e.g. ??). This method is especially powerful for A and F-type stars which have shorter main-sequence lifetimes between $\sim 1 - 5$ Gyr.

An approximate schematic of which tracer can produce the most precise age for different stars is shown in Figure ??. Overlap allows for robust age measurements from multiple tracers for some stars. Significant scatter is observed in the age indicators for a given age and mass, which can be quantized to produce a posterior probability distribution on age. The precision of a given age tracer depends on the mass and age of the star. Several tools have been developed to derive age posteriors from these tracers such as `BAFFLES` (?), `EAGLES` (?), `stardate` (?), and `gyro-interp` (?). We present median ages and 68% and 95% confidence intervals for 166 northern hemisphere accelerating stars based on these four age tracers below.

3. Sample, observations, and data reduction

We identify candidate exoplanet hosts by examining the *Hipparcos-Gaia* Catalog of Accelerations presented in ? created using the *Gaia* DR3 catalog. We

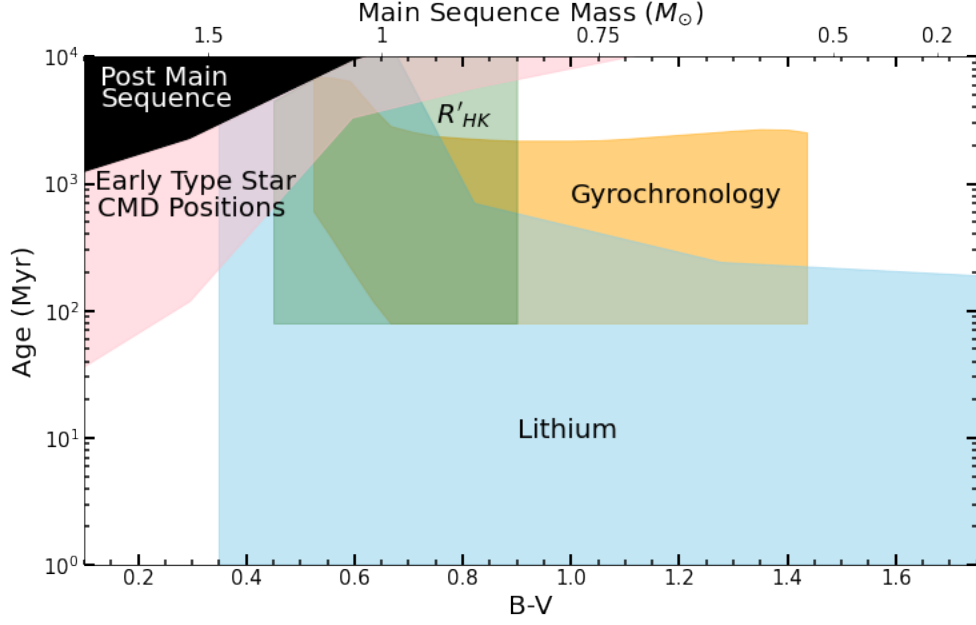


Fig. 2.1.— Age tracers used in this work for young, low-mass stars. This approximate schematic shows which combinations of age and mass are robustly covered by different age tracers. Each shaded region indicates where each tracer is most robust but does not include the regions where limits can be derived from each tracer (e.g. above the blue shaded region lithium is not expected to be detectable, but upper limits on lithium abundance can be used to derive lower limits on the age).

select stars with a $> 4\sigma$ proper motion anomaly, combining the significance of the three pairs of proper motions. We exclude known close binaries reported in the Washington Double Star Catalog (WDS) (?) or the Ninth Catalog of Spectroscopic Binary Orbits (SB9) (?) and stars with obvious companions in archival imaging data. Of the remaining stars, we reject those whose astrometric signal is consistent with a stellar companion between $0.1 - 1.0''$. We determine the allowed combinations of companion mass and orbital period using the rejection sampling

algorithm described in detail in ?. We compare simulated astrometric measurements from the simulated photocenter motion caused by an orbiting companion to the observed accelerations. We use the scan timings and scan angles of the *Hipparcos* and *Gaia* satellites to forward model the position and proper motion of the photocenter measured by the two missions. The photocenter motion is modeled as a combination of the linear motion of the system through space and the orbit of the primary star about the system barycenter. Figure ?? shows an example of the allowed mass-period parameter space for a star in our sample (HIP 669) based on the observed astrometric acceleration.

We limit our sample to stars observable from Apache Point Observatory (APO), i.e. stars with declinations $> -30^\circ$. We also limit the sample to stars within 50 pc to maximize the angular separation between the star and the potential companion to maximize the chance of the companion being directly imaged. We do not include stars that have archival radial velocities (e.g. from ESO/HARPS or Keck/HIRES). Figure ?? shows the distance, color, and magnitudes of our sample. The majority of our sample is between 30-50 pc with apparent G magnitudes between 6 and 11.

Age posteriors for lithium and R'_{HK} from BAFFLES (?) are calculated as a function of $B - V$. We adopt the $B - V$ values and associated errors that are reported in the *Hipparcos* catalog (?). Five stars in our sample (HIP 33368, HIP

46385, HIP 63661, HIP 69860, and HIP 81655) do not have a $B - V$ reported in the *Hipparcos* Catalog. An additional five stars (HIP 9067, HIP 111976, HIP 59953, HIP 1068, and HIP 1771) have very high uncertainties (~ 0.5 magnitudes) on their reported $B - V$ colors in the *Hipparcos* catalog. We derive $B - V$ for these targets by fitting a relation to the $B - V / G_{BP} - G_{RP}$ color-color plot (Figure ?? in Appendix). We use the $B - V$ values from the *Hipparcos* Catalog

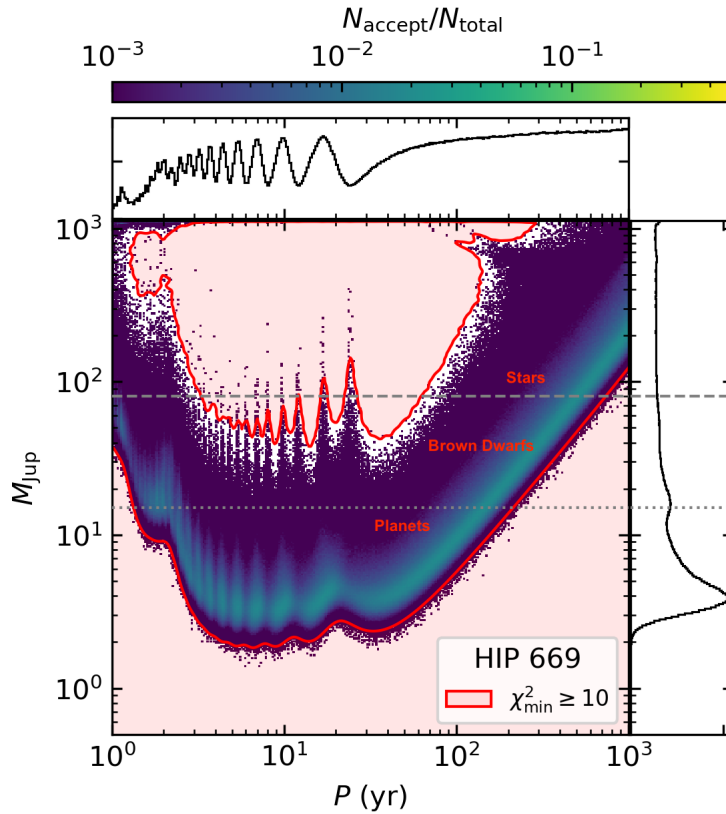


Fig. 3.1.— Results from rejection sampling of combined *Hipparcos* and *Gaia* DR3 astrometry show the possible companion mass-period combinations responsible for the observed acceleration of HIP 669. Candidate companions include stellar binaries, brown dwarfs, and wide-separation giant planets. The near-infrared contrast of these substellar companions depends on the age of the star.

(?) and the $G_{BP} - G_{RP}$ values from *Gaia* DR3 (?). We filter the *Gaia* catalog for all stellar entries that were cross-matched to *Hipparcos* by the *Gaia* team, merged the *Hipparcos*/*Tycho* data, we exclude variable stars, and then apply the following filters: $RUWE \leq 1.6$, $Gmag < 13$, $Gmag_{err} < 0.01$, $Hpmag_{err} < 0.1$, $(B - V)_{err} < 0.1$, $B - V \neq 0.0$, and $B - V < 2.0$.

We find the best-fitting relationship is

$$B - V = -0.102(G_{BP} - G_{RP})^4 + 0.226(G_{BP} - G_{RP})^3 + 0.017(G_{BP} - G_{RP})^2 + 0.688(G_{BP} - G_{RP}) + 0.002 \quad (3.1)$$

Based on the scatter in the relationship, we adopt a 0.1 magnitude uncertainty on the derived $B - V$ colors.

Gyrochronological age posteriors from **gyro-interp** depend on effective temperature (?). We determine effective temperatures for our sample by interpolating the grid from ? to the measured *Gaia* $G_{BP} - G_{RP}$ colors. To estimate the uncertainty on the effective temperature we find the range of effective temperatures corresponding to the range of $G_{BP} - G_{RP} \pm 0.1$.

To measure lithium equivalent widths and R'_{HK} , we obtain spectra using the Astrophysical Research Consortium Echelle Spectrograph (ARCES) on the APO 3.5 meter telescope. ARCES is a multi-use spectrograph with an average spectral resolution $R = \frac{\Delta\lambda}{\lambda} = 35,000$ covering 3,200 Å to 10,000 Å (?). We achieve SNRs

between ~ 20 for our faintest targets and ~ 200 for our brightest stars. The observing campaign began in October 2020 and is ongoing. We aim to observe each star at least three times over several months to years so that we can vet for binaries, which we will discuss in detail in future work. As of writing, we have observed each star at least once and 98 out of the 166 stars at least three times. We reduce the data using the python package, `pyvista`¹. Each spectrum is bias-corrected, extracted, flat-fielded, and continuum-corrected. Wavelength solutions are extracted from the ThAr taken nearest in time to the science frame and are then applied to the science spectrum.

We measure rotation periods from *TESS* light curves reduced with the SPOC pipeline (?). All light curves with cadences between 20-1800s are downloaded from the Mikulski Archive for Space Telescopes (MAST). We found light curves for 129 of our sample stars and identify periodic signals in the light curves of 34 of these 129 stars.

4. Lithium

We measure lithium 6707.79 Å equivalent widths from ARCES spectra so that we can use them and the python package `BAFFLES` (?) to derive age posteriors. First, we bring the spectrum to the approximate rest frame using the *Gaia* DR3

¹<https://pyvista.readthedocs.io/>

radial velocity measurement (?) for the star and correct for the Earth’s motion by applying a barycentric correction. To find the lithium 6707.79 Å feature and measure its equivalent width, we fit a double Gaussian with a slope to the spectrum with the functional form

$$f_{mod} = (m(\lambda - 6707.0) + b)(C_1 e^{-0.5(6707.79 - RV_s - \lambda)^2 / \sigma^2} + C_2 e^{-0.5(\lambda_{ref} - RV_s - \lambda)^2 / \sigma^2}) \quad (4.2)$$

where λ is the wavelength array, m and b are the slope and intercept of the continuum; λ_{ref} is the central wavelength of some reference feature; C_1 is the depth of the lithium feature; C_2 is the depth of the reference line; σ represents the standard deviation of both Gaussians; RV_s represents an additional radial velocity shift to the *Gaia* DR 3 measurement. We include this additional RV offset term in our fit to account for potential orbiting companions or any radial velocity drift in ARCES. For most stars, we adopt $\lambda_{ref} = 6705.1$ Å, a prominent Fe line. For lower mass stars where the 6705.1 Å Fe line is not present, we adopt $\lambda_{ref} = 6717.8$ Å, a prominent Ca line. The reference line prevents the fit from selecting a value for the radial velocity, RV_s that shifts another line (or noise trough) to the location of the lithium line. This also prevents the fit from adjusting the width of the lithium feature to match a noise trough. This is especially useful in cases where the lithium line is weak or not detectable. When $\lambda_{ref} = 6705.1$ Å we use the region of spectra from 6704 Å to 6710 Å. When $\lambda_{ref} = 6717.8$ Å we use the region

of the spectrum from 6706 Å to 6720 Å. Both of these regions fall on a single ARCES order.

Since the 6707.79 Å feature is a blend of the ${}^6\text{Li}$ and ${}^7\text{Li}$ doublets, we find the effective center of the entire feature by fitting the double Gaussian to a narrow region of spectrum from 6704 Å to 6710 Å for several stars where $\lambda_{ref} = 6705.1$ Å such that

$$f_{mod} = (m(\lambda - 6707.0) + b)(C_1 e^{-0.5(6707.79+s-RV_s-\lambda)^2/\sigma^2} + C_2 e^{-0.5(6705.105-RV_s-\lambda)^2/\sigma^2}) \quad (4.3)$$

Here, s is the deviation of the center of the lithium feature from 6707.79 Å.

We use the python package *emcee* (?) to run Markov Chain Monte Carlo (MCMC) fits on several spectra and found that s for the best fit was consistently 0.0119 Å regardless of spectral type, so we adopt 6707.819 Å as the effective center of the lithium feature.

With Equation 3 and *emcee*, we run MCMC to find the best-fitting parameters and estimates of their uncertainties for each spectrum. The uncertainties in each pixel of the spectrum are assumed to follow a Poisson distribution and include estimates of the gain and read noise.

First, we run a preliminary MCMC with 100 walkers and 10,000 steps to find

good starting positions. Then, a second round of MCMC is run from these starting positions, with 100 walkers, for 50,000 steps with a 2,000 step burn-in, which was sufficient to achieve convergence for most stars. 16 stars with non-detections required 100,000 steps with a 10,000 step burn-in, and 2 additional stars required 400,000 steps with a 200,000 step burn-in. We assessed convergence both by visual inspection and by comparing the posteriors at the start and end of each chain. We also compute the integrated autocorrelation time (IAT) of our chains, which has a maximum value of 1.2; we therefore estimate that there are at least 35,000 independent samples of the posterior, consistent with the start and end of each chain being independent. Comparing the median of each parameter at the first 10% of each chain to the final 10% we find the largest fractional difference to be 9%. A similar test comparing the standard deviation from the first and last 10% yields a maximum fractional difference of 6%. These results are consistent with convergence. To calculate the lithium equivalent width, we integrate a Gaussian with parameters from the MCMC chains

$$\Sigma_{Li} = \sqrt{2\pi}C_1\sigma \quad (4.4)$$

which we converted to an equivalent width by scaling the area by the height of the nearby continuum, H

$$EW_{Li} = \frac{\sqrt{2\pi}C_1\sigma}{H} \quad (4.5)$$

To find the height of the nearby continuum, we evaluate a line defined by the slope (m) and intercept (b) of the fit at 6707.819 Å (the center of the lithium feature). We use our chains to compute the posterior on equivalent width and report the median. We take the standard deviation of the equivalent width posterior from the converged chain as our uncertainty.

For stars that do not have a clear lithium feature, we use the same MCMC fitting method to calculate upper limits on the lithium equivalent width. For upper limits, C_1 in Equation 3 approaches zero, and σ and RV_s are set almost entirely by the reference line. We identify upper limits by measuring the skew of the equivalent widths' distribution. If the skew exceeds 0.225 or the mean of the distribution is below 5 mÅ, we flag the spectrum as a candidate upper limit. We confirm our upper limit selection by visually inspecting the spectrum. We calculate the 1, 2, and 3σ upper limits by calculating the 68th, 95th, and 99.7th percentiles of the cumulative distribution function. We over-plot Gaussians with σ determined by the reference line and depths set by the 1, 2, and 3σ upper limits to visually assess the corresponding upper limit on the equivalent width. We report the 2σ upper limits.

We calculate the equivalent width for each star using both spectral orders that include the lithium 6707.8 Å feature. We adopt the equivalent width measurement or upper limit from the order/observation combination that yields the best fit to

the data. This is generally the segment with the highest signal-to-noise ratio and no cosmic rays. We confirm our selection by visually assessing the fits. Each fit typically returns similar measurements of the lithium equivalent width, with error bars scaling with SNR as we would expect. Figure ?? shows examples of a lithium detection and a lithium upper limit determined from this process.

For stars with large amounts of lithium or high rotational velocities, the lithium 6707.8 Å feature can be blended with the 6707.4 Å Fe line. Past work has introduced a correction factor to equivalent widths to subtract the contribution of the iron line from the blended equivalent width. For example ? calculate iron 6707.4 Å equivalent widths as a function of $B - V$, and subtract this from their blended equivalent widths. Similar methods are used by ??????. To assess the degree to which including an iron blend correction improves our lithium equivalent width measurements, we compiled literature values where available for our target stars and S-value calibration targets (discussed below) from ?. In Figure ??, we compare our measured lithium equivalent widths with and without the ? correction to these literature values for 12 stars. We find that our method with no correction applied yields more consistent equivalent widths compared to these literature values. Applying the ? correction systematically under-predicts the literature values. We attribute this to our method of fitting a Gaussian to the full absorption profile and taking the integral of the Gaussian, rather than summing fluxes in a specific wavelength range directly from the spectrum. As a result, since

the asymmetric wing from the iron line introduces a negligible effect on the final equivalent width, we do not apply a correction to our measured equivalent widths.

To ensure there are no systematic offsets in our lithium fitting technique, we inject a Gaussian of a given equivalent width into an ARCES spectrum with no detectable lithium. Then we feed the injected spectrum through our fitting routine and recover the equivalent width. We conduct this test on five spectra with moderate to high signal-to-noise corresponding to stars at a range of spectral types. For each star, we repeat the injection/recovery test for 13 different values of injected lithium equivalent widths from 5-100 mÅ. Figure ?? compares the recovered equivalent widths to the injected lines, which agree to $\lesssim 1$ mÅ, well within the reported error bars for each measurement. The main source of this disagreement is noise in each spectrum with the noise being on average slightly above or slightly below the continuum at the location of the lithium line, resulting in a slight underestimate or overestimate of the equivalent width, respectively.

We calculate age posteriors from lithium equivalent widths and uncertainties for our target stars using Bayesian Ages For Field LowEr-mass Stars (BAFFLES) (?). BAFFLES utilizes a Bayesian framework that is calibrated on stars in clusters and associations of known ages. Functional forms were fit both to lithium equivalent width as a function of $B - V$ for each calibration cluster, and the intrinsic scatter in equivalent width. Final age posteriors represent a combination of the

astrophysical scatter in the age/equivalent width relation and the measurement error (?). **BAFFLES** uses a uniform star formation history as the age prior. The uncertainty on age is largely driven by the observed astrophysical scatter in lithium equivalent width at a constant age, with a much smaller contribution from the measurement uncertainty for a single star. **BAFFLES** is calibrated to calculate age posteriors from lithium for stars with $B - V$ between 0.35 and 1.9 from both measurements and upper limits. Figure ?? shows an example **BAFFLES** posterior for HIP 41277, where an equivalent width of 125 ± 5 mÅ translates to an age posterior of 174^{+79}_{-77} Myr. We detect lithium in 26 stars, and determine upper limits for 140 stars; of the full sample, 159 stars are in the **BAFFLES** $B - V$ range. We report $B - V$, lithium equivalent width measurements (or upper limits), and median ages and confidence intervals from **BAFFLES** (where applicable) for our sample in Table ??.

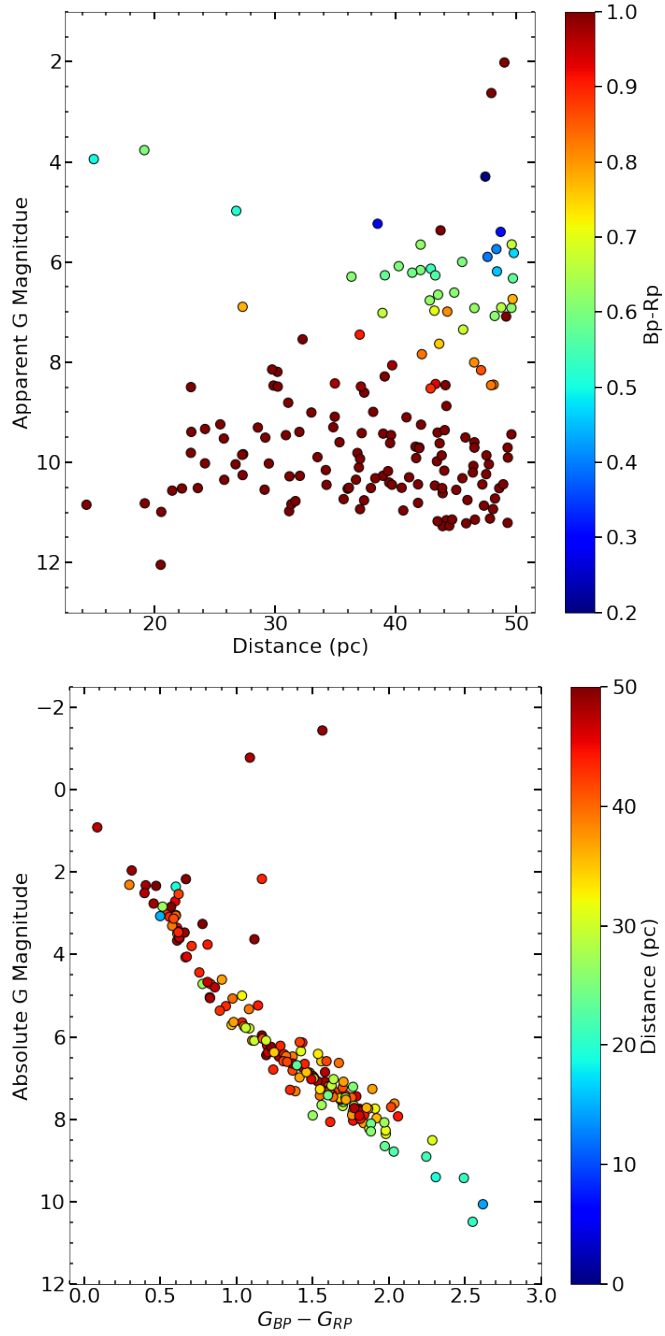


Fig. 3.2.— **Left:** Apparent *Gaia* G magnitude of the accelerating stars as a function of distance. **Right:** Color-magnitude diagram of the sample. The median of our sample is $G_{BP} - G_{RP} = 1.4$ ($\sim$ K5), 41.5 pc, and $G = 9.6$. Our sample covers a wide range of spectral types, so we require a variety of age tracers to derive ages for each star.

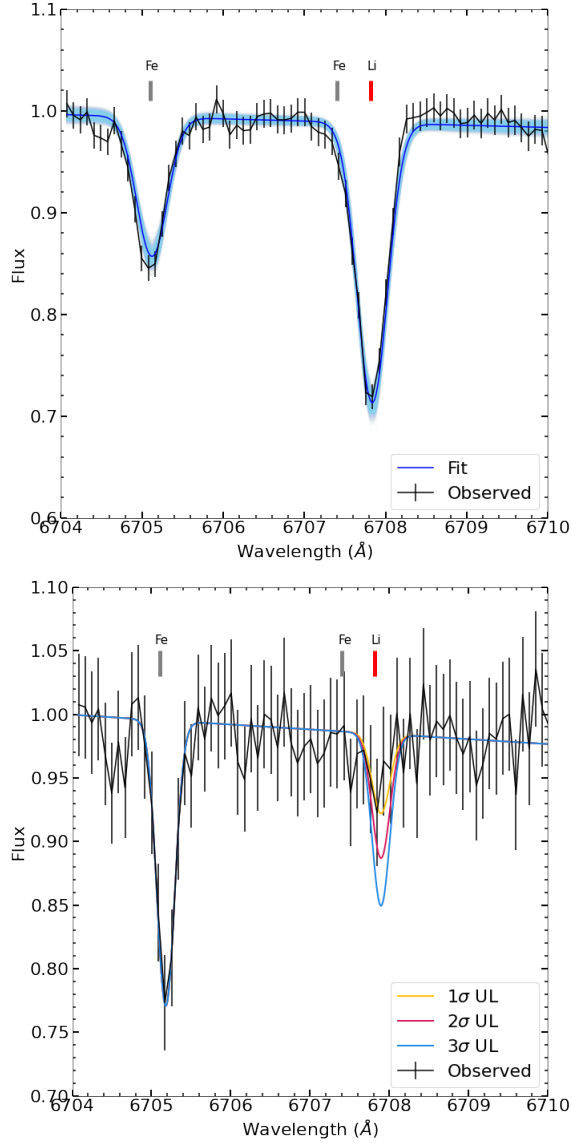


Fig. 4.1.— Our double Gaussian fitting technique allows us to robustly measure lithium equivalent widths and, in cases where lithium is not detected, place upper limits on the lithium equivalent widths. **Left:** A spectrum of HIP 41277 ($SNR \sim 75$) covers both the Fe 6705.1 Å line (used as a reference line) and the Li 6707.79 Å feature. Black points with error bars are the APO/ARCES data, the dark blue line represents the highest likelihood fit from the MCMC, and the light blue lines are random draws from the posterior. We find a lithium equivalent width for HIP 41277 of 125 ± 5 mÅ. **Right:** An example spectrum ($SNR \sim 30$) of an upper limit for lithium, from HIP 110663. Data are again black points with error bars, while yellow, red, and blue represent 1, 2, and 3 σ upper limits. We find a 2 σ upper limit of 30 mÅ for HIP 110663, which we use to set a lower limit on the age of the star.

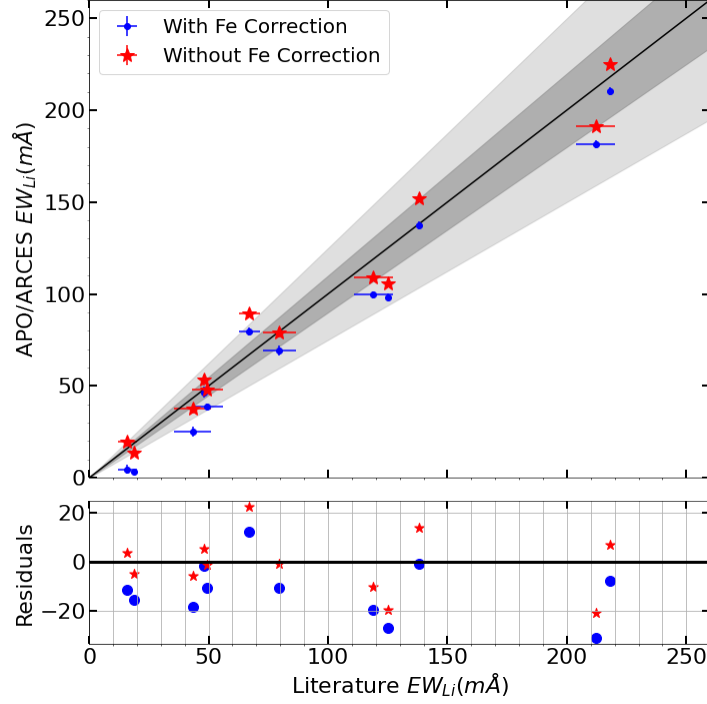


Fig. 4.2.— Comparing our lithium equivalent width measurements to results from the literature for 12 stars (red stars) shows good agreement. The black line shows the 1-to-1 relation, with $\pm 10\%$ in dark gray and $\pm 25\%$ in light gray. We find that applying a correction for the 6707.4 Å iron line (?) results in a systematic underestimate of the literature abundances (blue points). We conclude that our Gaussian fitting technique is robust to contamination from the 6707.4 Å iron line, so we do not include an additional correction to our reported equivalent width values. We take literature lithium equivalent widths for HIP 104075, HIP 105232, HIP 113905, HIP 35628, HIP 97779 from ?, HIP 115147, HIP 43410, HIP 52462, HIP 71631 from ?, HIP 42438 from ?, and HIP 43726 from ?.

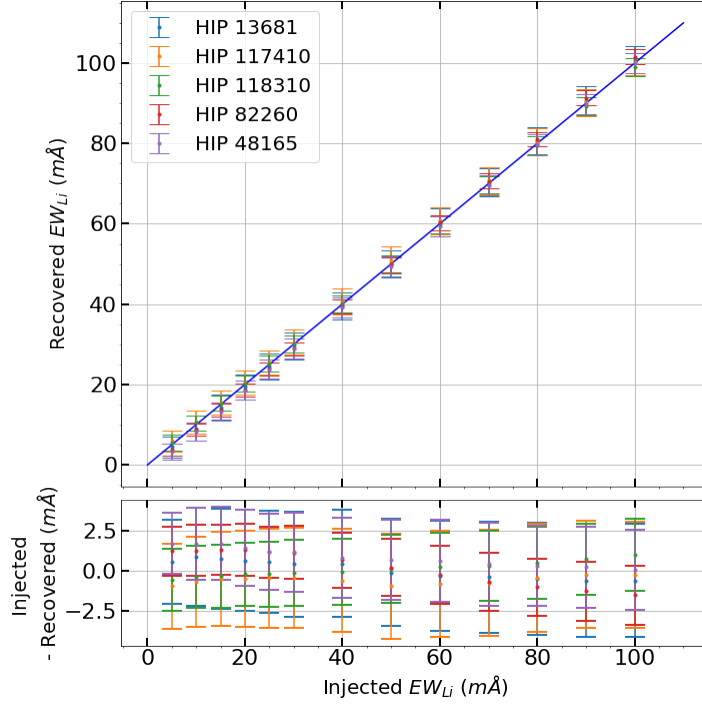


Fig. 4.3.— We confirm that our lithium fitting routine reliably recovers accurate equivalent widths with injection/recovery tests on 5 stars. We chose these stars since they had no detectable lithium, and because they cover a range of SNR and spectral types. The recovered equivalent widths are all within $\lesssim 1 m\text{\AA}$ of the injected value, and consistent with reported error bars.

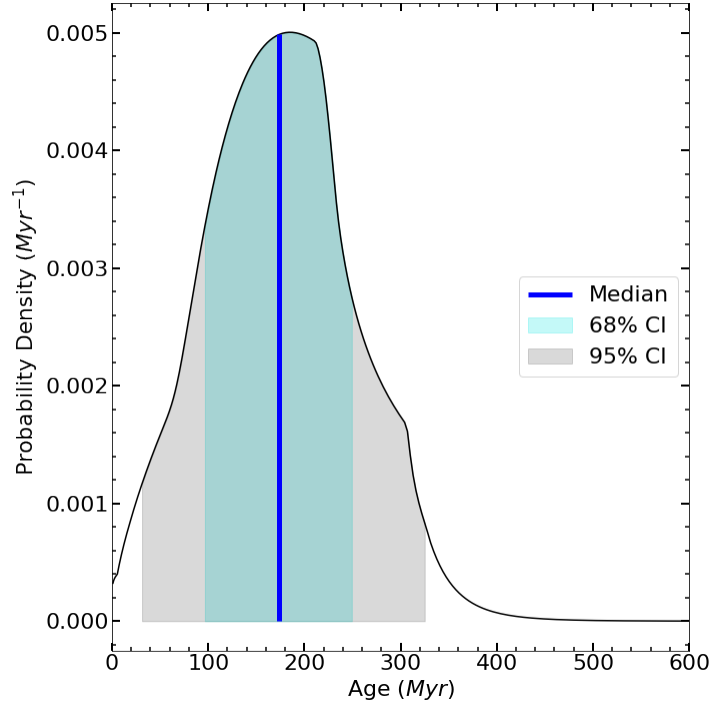


Fig. 4.4.— An example of an age posterior for one star in our sample, HIP 41277, using BAFFLES (?). From an adopted $B-V$ of 1.03 ± 0.015 and a lithium equivalent width of 125 ± 5 mÅ, we derive an age posterior with a median age of 174 Myr and 68% confidence interval between 97 and 253 Myr. Lithium measurements and median ages, and 68% and 95% confidence intervals for the full sample are reported in Table ??.

5. Calcium Emission

We also use **BAFFLES** to derive age posteriors from calcium H (3968.47 Å) and K (3933.67 Å) emission (R'_{HK}) for stars with spectral types between F5 and K2. R'_{HK} is derived from the S-value index, which is the ratio of calcium emission flux to nearby continuum. S-values were initially measured photometrically following the methods of the Mt. Wilson Ca II H & K survey that ran from 1966 to 1983 (?). The emission flux is measured through two triangular bandpasses, and the continuum flux is measured through two rectangular bandpasses on either side of the emission cores, as shown in Figure ??.

Subsequently, the photometric S-value index is most commonly measured from spectra with the S-value index calibrated for each instrument. To calibrate APO/ARCES, we observe 99 stars with published S-values from ?, which compiled published S-values from the literature including ?, ?, and ?. We follow a prescription similar to that in ? with some modifications to calibrate the APO S-values. We use a 1-parameter fit (compared to the 4-parameter fit of ?) with the form

$$S = a \left(\frac{H + K}{R + V} \right) \quad (5.6)$$

H and K are the summed counts from convolving the spectrum with the triangular transmission profiles, while R and V are the summed counts from convolving the continuum with the rectangular transmission profiles (taken from ?).

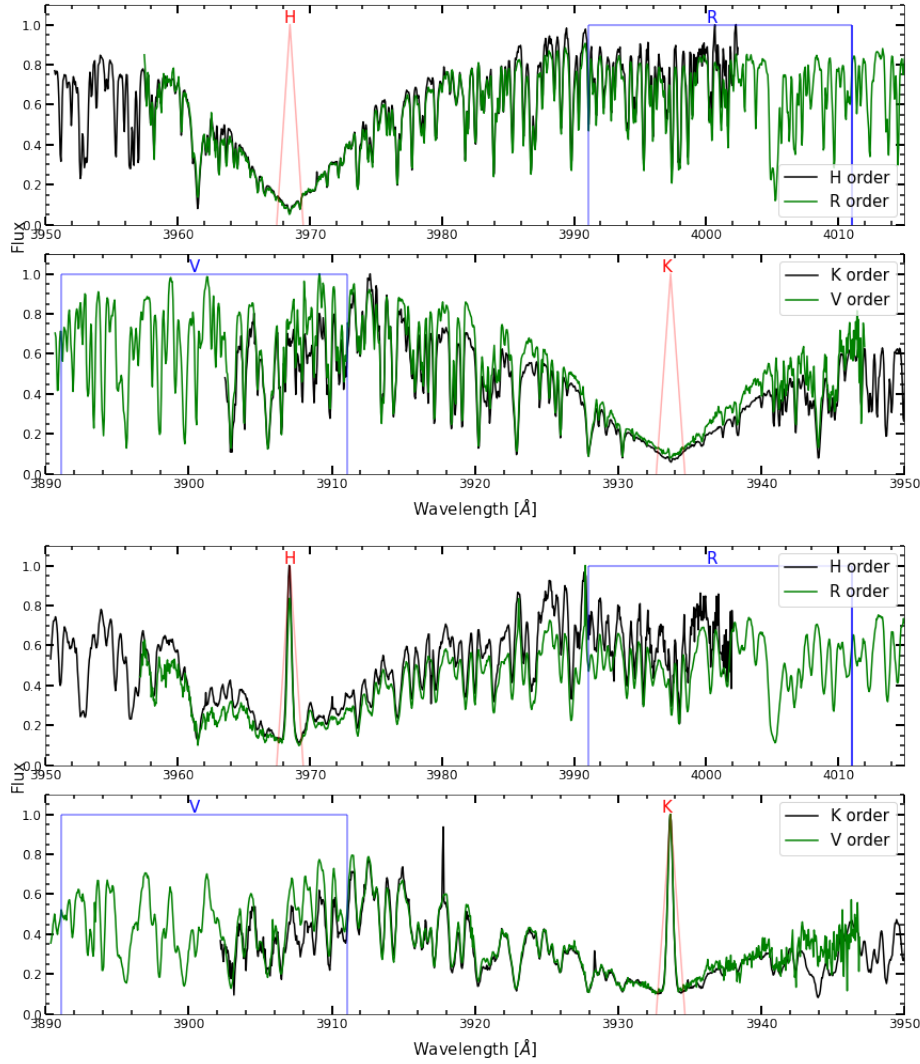


Fig. 5.1.— The S-value is measured from spectra using triangular transmission profiles (red lines) centered on the Ca II H and K emission lines, and rectangular profiles (blue lines) to measure the nearby continuum level. The top two panels show an example of a low-activity star with no significant H and K emission (HIP 102040), and the bottom two panels show HIP 115147, a star with strong H and K emission.

We shift the spectra to the rest frame so the calcium emission cores or the centers of the absorption lines are centered in the narrow triangular band passes.

Using the $G_{BP} - G_{RP}$ to $B - V$ conversion described in Section ??, and the

grid from ? we find effective temperatures, masses, and radii for our targets, assuming they are on the main sequence. We find the synthetic spectrum with solar metallicity from ? with the closest effective temperature and surface gravity. We convolve the synthetic spectrum with a Gaussian to match the resolution of ARCES. Then, we shift the ARCES spectra to the rest frame by cross correlating the observed spectrum with the theoretical spectral template.

Literature S-values are often not given with error bars, and generally variation over the star’s activity cycle is expected to be larger than measurement errors. Given the lack of reliable error bars we do not attempt to derive a posterior probability distribution on the fit parameter. Instead, we assume error bars of unity for each measurement, and find the value of a in Equation ?? that minimizes the χ^2 . We compute this minimum value over a uniform grid for a with 10^6 elements between 0 and 40, finding a best-fit value of $a = 18.3$, which results in a $\sim 15\%$ scatter on the relationship between APO S-values and the literature S-values. The fit is shown in Figure ??, and we attribute this $\sim 15\%$ scatter mainly to variable emission over a stellar cycle.

We investigated more complicated fits, such as including additional parameters to weight the H and K bands or the R and V bands differently, or attempting to further correct the continuum by comparing to theoretical spectra, or performing different fits for different spectral type bins. None of these more complicated

methods significantly improved the scatter, and some resulted in systematic offsets from the fit at larger S-values. As a result, we adopt the more straightforward approach outlined above.

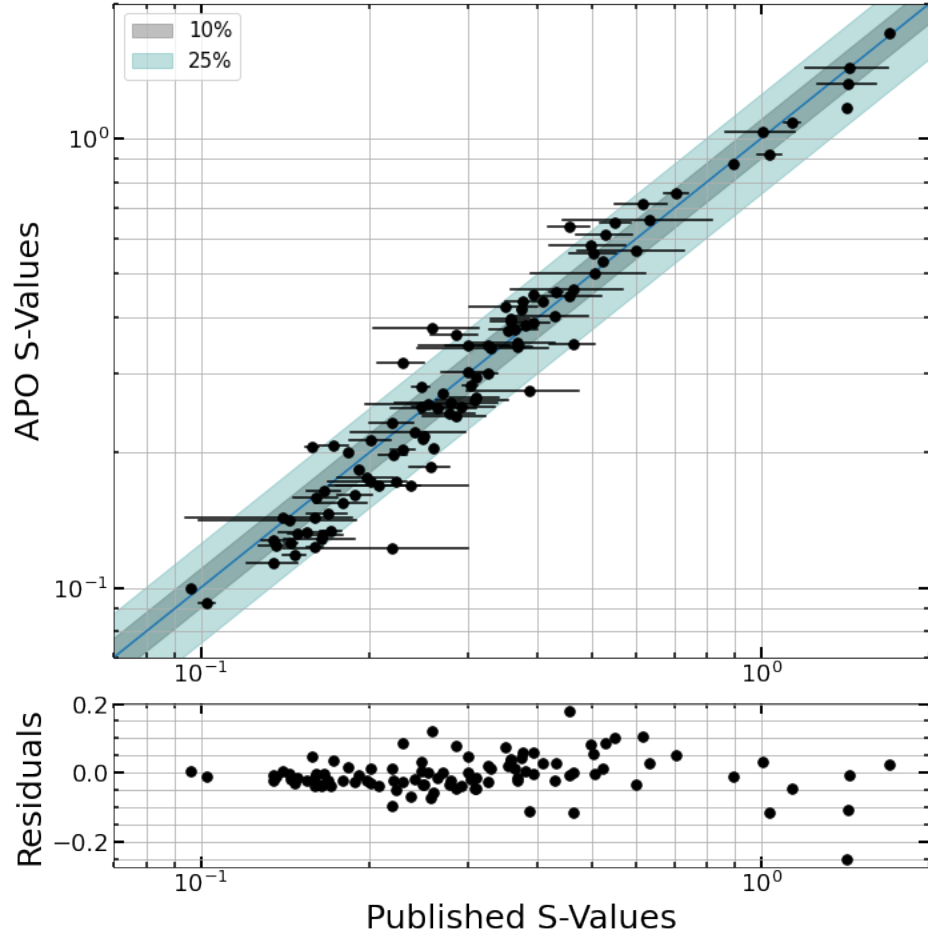


Fig. 5.2.— We use a 1 parameter fit to calibrate chromospheric S-values from APO/ARCES to values reported in the literature. We observe a $\sim 15\%$ scatter in the relationship, which we largely attribute to the intrinsic variability in chromospheric activity for each star. Error bars represent the maximum and minimum S-value reported in the literature for each star. We report S-values and $\log(R'_{HK})$ for all the accelerating stars in Table ??.

The transformation from S-value to R'_{HK} is

$$R'_{HK} = 1.34 \times 10^{-4} C_{Cf} S - R_{phot} \quad (5.7)$$

where C_{Cf} is the color correction factor, a third-order polynomial in $B-V$ (?), and R_{phot} is another third-order polynomial in $B-V$ that corrects for the photospheric contribution to the emission in the Ca H and K lines (?). S and R'_{HK} are given

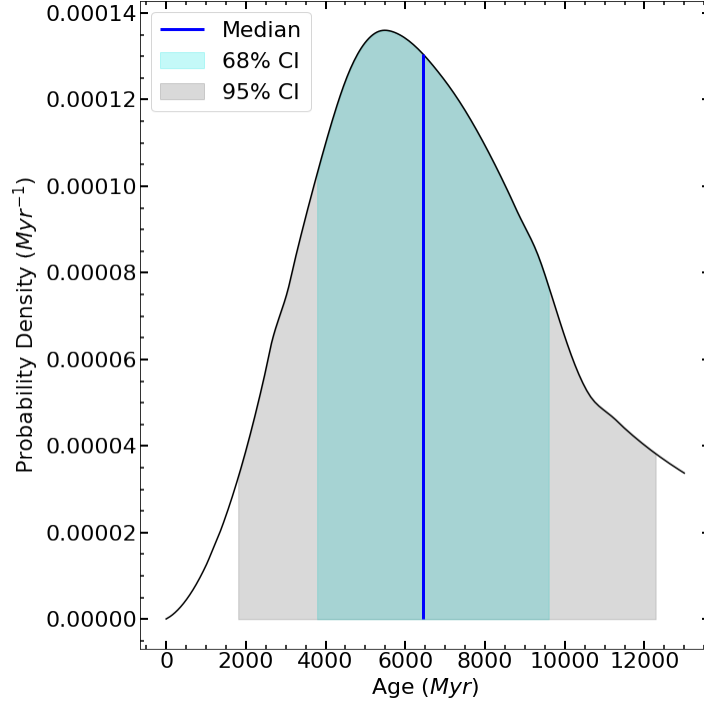


Fig. 5.3.— An example age posterior for HIP 669, using our APO/ARCES measurement and BAFFLES (?). For this relatively inactive star, with $\log(R'_{HK}) = -4.88$ and $B - V = 0.62 \pm 0.015$, we derive an age posterior with a median age of 6.5 Gyr and 68% confidence interval between 3.8 and 9.7 Gyr.

for each of our target stars in Table ??, as well as median ages and 68% and 95%

confidence intervals for the 34 stars between F5 and K2 derived using **BAFFLES** (?). Similar to lithium, **BAFFLES** calculates age posteriors from R'_{HK} values in this spectral type range based on empirical fits to R'_{HK} as a function of age and the intrinsic scatter as observed in clusters and associations of known ages. Figure ?? shows an example age posterior for HIP 669, with a $\log(R'_{HK})$ of -4.88 that gives an age posterior of $6.5^{+3.2}_{-2.7}$ Gyr.

6. Gyrochronology

We calculate gyrochronological ages using the python package **gyro-interp** which computes age posteriors by interpolating between rotation sequence models for open clusters with well-defined ages (?). **gyro-interp** is calibrated for stars with $3800K < T_{eff} < 6200K$. We measure rotation periods from *TESS* light curves processed with the SPOC pipeline (?), which were available for 129 of our target stars. We visually inspect each light curve and its corresponding Lomb-Scargle periodogram (??) to make an initial estimate for the rotation period. Then, to further validate and refine the selected period, we use the non-linear least squares fitting python package *LMFIT* (?) to fit a sinusoid to the light curve. We run *LMFIT* to find the sinusoid’s amplitude, phase, vertical offset, and period. We limit the period to a narrow range around the initial estimate to prevent the fit from getting stuck at a local minimum at a harmonic of the

true period. We do not use MCMC or another sampling algorithm to find the rotation period because a sinusoid is only an approximation of spot modulation and because photometric measurements are correlated over small time spans.

To ensure we select an actual rotation signal we compare our adopted period to peaks in the corrected (PDCSAP) and the uncorrected (SAP) periodograms. The period must appear as a local maximum in both periodograms to be accepted. We do not require the period to be the global maximum of either periodogram.

We differentiate between “flat” light curves and robust variability signals by comparing χ^2 for a flat line and the best-fitting sinusoid. Based on visual inspection of light curves from our target stars, we identify light curves with $\Delta\chi^2 \gtrsim 500$ as having a variability signal, and classify all other light curves as flat. Regardless of their $\Delta\chi^2$, 25 stars have variability amplitudes $\lesssim 0.1\%$ and are taken to be flat. We also visually confirm variability in each light curve. We do not measure rotation periods for 8 stars that share a *TESS* pixel with another star that is $\gtrsim 2.5\%$ as bright as the target star in the *Gaia* G band. We also visually inspect the target-pixel files (TPFs) for bright stars on the edges of nearby pixels.

For stars observed in multiple *TESS* sectors, we adopt the period from the light curve with the fewest and/or smallest down-link gaps, which we determine by visual inspection. If more than one curve has similar amounts of data gaps, we select the light curve with the highest signal-to-noise. We use the other sectors to

further validate our adopted rotation periods. Some stars are not variable in some sectors but are highly variable in others. We attribute this to spot evolution, such that spots are present in one sector but not present in another. We adopt the rotation period from the sector displaying variability.

Measuring rotation period uncertainties can be challenging, and there are several different methods commonly used. For example, [Barnes et al. \(2016\)](#) estimate rotation uncertainties by the width of the periodogram peak, while [Barnes \(2017\)](#) use the width of the autocorrelation peak. We estimate uncertainties on the measured rotation periods with a similar method, by fitting a Gaussian to the exponentiated Lomb-Scargle periodogram peak, which is proportional to the likelihood of a single-sinusoid model at each frequency ([Barnes et al. 2016](#)). [Barnes \(2017\)](#) cautions that this method does not take into account the possibility of having selected the wrong periodogram peak. Therefore, we validate our period selection in two ways: visual inspection and false alarm probability (FAP). For all our measured periods, the largest FAP we calculate is 0.5%, and all the light curves cover more than one period making visual identification of the period relatively robust. We propagate the σ of the best fitting Gaussian in frequency space to period, such that

$$\sigma_P = \sigma_f P^2 \tag{6.8}$$

We use `gyro-interp` (?) to turn these rotation periods and uncertainties into age posteriors, with an example gyrochronological age posterior shown in Figure ?? . `gyro-interp` generates age posteriors from rotational periods in a similar manner to how `BAFFLES` generates posteriors from lithium and calcium measurements. A reference sample of stars in clusters and associations of known ages was used to find the distribution in rotation period as a function of age and temperature, as well as the astrophysical scatter. The generated age posterior is then a combination of the rotation period and errors and the astrophysical scatter. As with lithium, the dominant source of uncertainty in the final posterior is astrophysical scatter rather than measurement uncertainty on the period.

6.1. Complex light curves

Several *TESS* light curves of our targets do not exhibit straightforward rotation signals. In this section, we briefly describe the different types of variability we observe (e.g. ?).

Double periods - HIP 117410 shows two distinct periods in its light curves, but is not on obviously contaminated pixels. We interpret this as a rotation signal of two stars on the same pixels. To determine each period, we visually assess each light curve and make an initial estimate for each period. Then, we fit a double

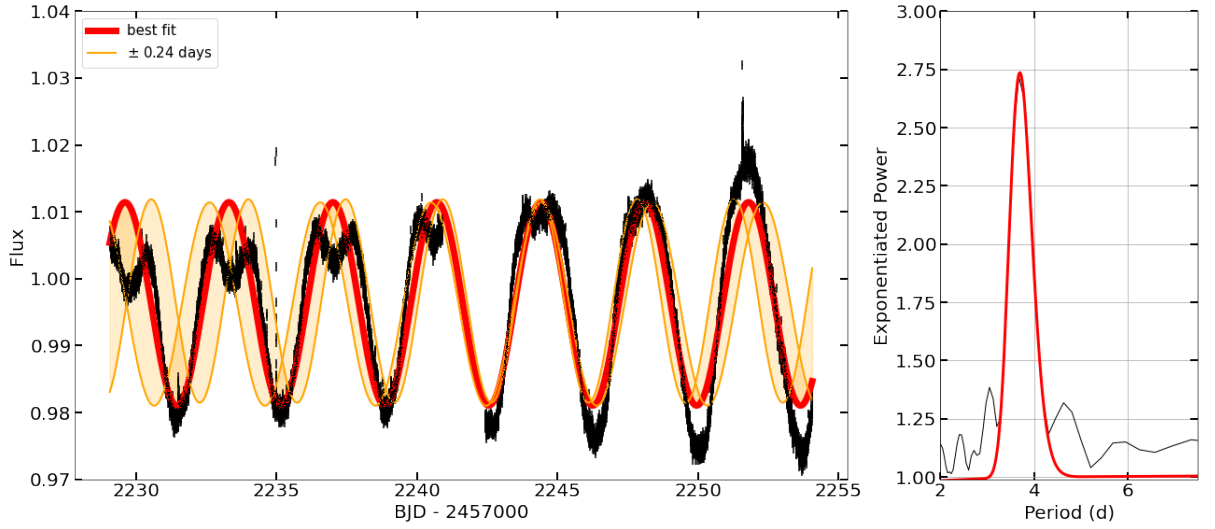


Fig. 6.1.— We measure rotation periods for the accelerating stars by analyzing their periodograms and fitting sinusoids to their lightcurves. The *TESS* light curve for HIP 41277 is shown in black with the best fitting sinusoid with a period of 3.7 days overplotted in red. The orange curves represent the best-fitting period adjusted by the uncertainty measured from the width of the exponentiated periodogram peak. The right-hand panel shows the exponentiated Lomb-Scargle periodogram with a significant peak at ~ 3.7 days and the best fitting Gaussian.

sinusoid to the flux of the form,

$$F = A_1 \sin(2\pi f_1 t + \phi_1) + A_2 \sin(2\pi f_2 t + \phi_2) \quad (6.9)$$

to each light curve using *LMFIT*. f_1 and f_2 are the frequencies, A_1 and A_2 are the amplitudes, and ϕ_1 and ϕ_2 are the phase shifts of each of the component sinusoids.

We find the best fitting periods for each component to be 10.65 ± 1.07 days and 0.69 ± 0.07 days. The best-fitting double sinusoid for HIP 117410 is shown in Figure ??.

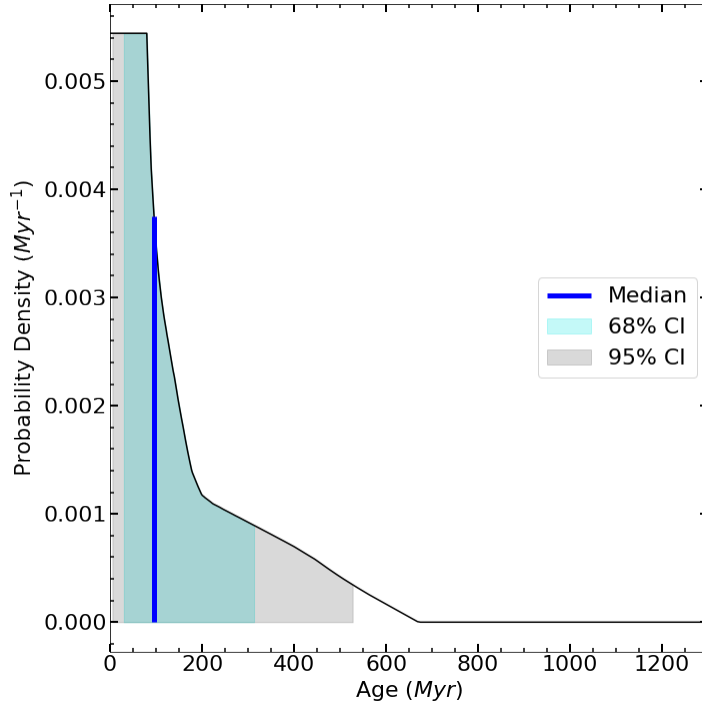


Fig. 6.2.— We calculate age posteriors from rotation periods using `gyro-interp` (?). The age posterior for HIP 41277 with an adopted T_{eff} of 4551 ± 195 K and a measured rotation period of 3.70 ± 0.24 days has a median age of 96 Myr and a 68% confidence interval between 30 and 316 Myr.

HIP 117410 (WDS 23484-1259) is a $\sim 1''$ double star with a flux ratio of ~ 4 in V band (?), so this pair is most likely the source of the two signals. This pair was not flagged as a contaminated pixel due to the absence of a *Gaia* G band measurement for the secondary. Early analysis of the APO/ARCES radial velocities of HIP 117410 (to be discussed in detail in a future paper) suggests that the system is not a short-period stellar binary.

We cannot determine which period corresponds to which star, or directly measure $B - V$ for the secondary with the current data. Instead, we derive an

age posterior for each period with the B-V of the primary, then marginalize over each star by summing the two posteriors and normalizing the resulting posterior so that it integrates to 1 (Table ??).

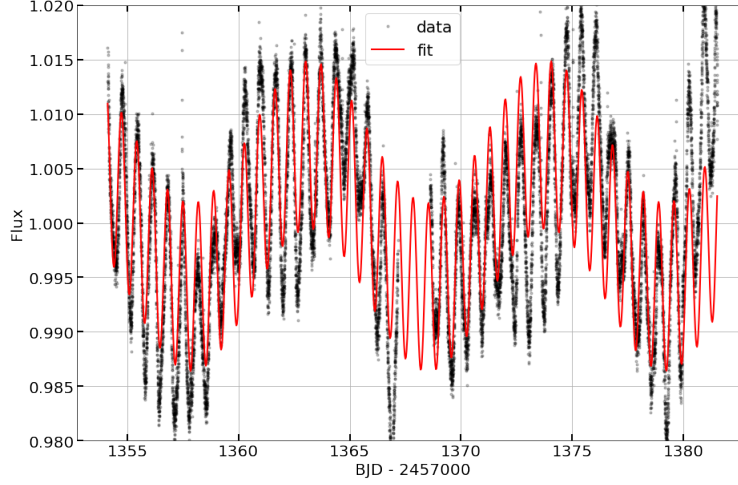


Fig. 6.3.— We fit a two-component sinusoid model (red) to a light curve (black) for HIP 117410. We find the periods of the two components to be 10.65 ± 1.07 days and 0.69 ± 0.07 days

P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)
10.65 ± 1.07	415.99	214.20–967.37	146.59–11492.34
0.69 ± 0.07	105.19	32.77–337.67	5.56–542.49
Combined	254.07	64.24–625.41	10.11–1400.23

Table 6.1: The age posteriors for HIP 117410 for both periods and the combined posteriors are listed. We adopt the combined age posterior for this system.

Eclipsing Binaries - We identify HIP 16247, HIP 45731, and HIP 49577 as eclipsing binaries. HIP 45731 and HIP 16247 were previously analyzed in ? and

?, respectively.

We calculate the orbital period and rotational periods separately for each of these three systems. To find the orbital period we phase-fold the light curve based on the highest periodogram peak, verifying that the eclipses overlap, and the primary and secondary eclipses are not repeated in the folded light curve. If the phase folded curve has too many eclipses or the eclipses do not overlap, we try the next harmonic. We estimate the uncertainty of the orbital period by altering the period until the eclipses no longer overlap in the phase-folded light curve. To measure the underlying rotational period, we mask out the eclipses and then use the method described above for single stars. Table ?? gives the orbital periods (P_o) and rotational periods (P_r) for each system as well as approximate depths of the primary (D_p) and secondary (D_s) eclipses. Figure ?? shows the light curve for the eclipsing binary HIP 16247 with the best fitting sinusoid to the underlying rotation plotted in red. The data and the sinusoid fit are out of phase with each other between $1420 < BJD - 2457000 < 1425$. We attribute this to the fact that a simple sinusoid does not account for spot evolution and changing amplitudes, which can shift the phase of the periodicity, but a sinusoid is sufficient for capturing the periodicity of the underlying rotation rate.

In close binaries, the stars can be spun up by tidal forces, so that the rotation period reflects tidal evolution rather than spin-down following the empirical

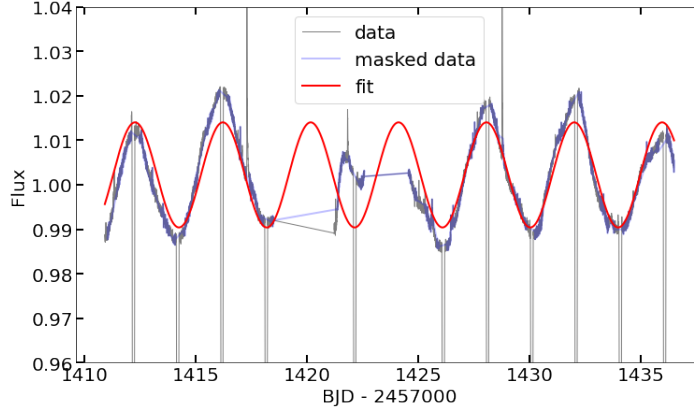


Fig. 6.4.— We fit a sinusoid to the underlying rotation signal of an eclipsing binary system. The light curve for HIP 16247 is shown in gray. The masked data with the eclipses removed is shown in purple, and the best-fitting sinusoid is shown in red. This star is tidally locked with $P_{orb} \approx P_{rot} \approx 3.98$ days.

Target	WDS	P_o (d)	D_p	D_s	P_r (d)
HIP 16247	03294-2406	3.98 ± 0.004	$\sim 28\%$	$\sim 22\%$	3.95 ± 0.4
HIP 45731	09194+6203	2.92 ± 0.003	$\sim 15\%$	$\sim 14\%$	2.92 ± 0.3
HIP 49577	—	16.85 ± 0.17	$\sim 24\%$	$\sim 10\%$	7.2 ± 0.72

Table 6.2: The estimated parameters of the three eclipsing binaries in our sample.

gyrochronology relations. For HIP 1627 and HIP 45731, the rotation periods and orbital periods we measured are approximately the same suggesting that both systems are tidally locked. Therefore, we do not derive gyrochronological ages for these systems. For HIP 49577, the rotation rate we derived is significantly different than the orbital period we derived. From the rotation period and $B - V$ of the primary star, we measure a median age of 211 Myr and 68% confidence interval of 105 – 422 Myr. However, while this system is not tidally locked, we cannot rule out tidal forces affecting the rotational evolution of the system.

Other Variable Stars - We identify several stars that exhibit variability that does not resemble the rotational modulation of star spots. We use the examples in ? to visually classify these stars as stochastic or beat pattern variables. HIP 11090 and HIP 59767 both display a beat pattern in their light curves. HIP 21066, HIP 41274, HIP 22361, and HIP 59199 exhibit stochastic variability. An example of each is shown in Figure ??. Along with not displaying clear rotation signals, most of these stars are outside the `gyro-interp` temperature range and

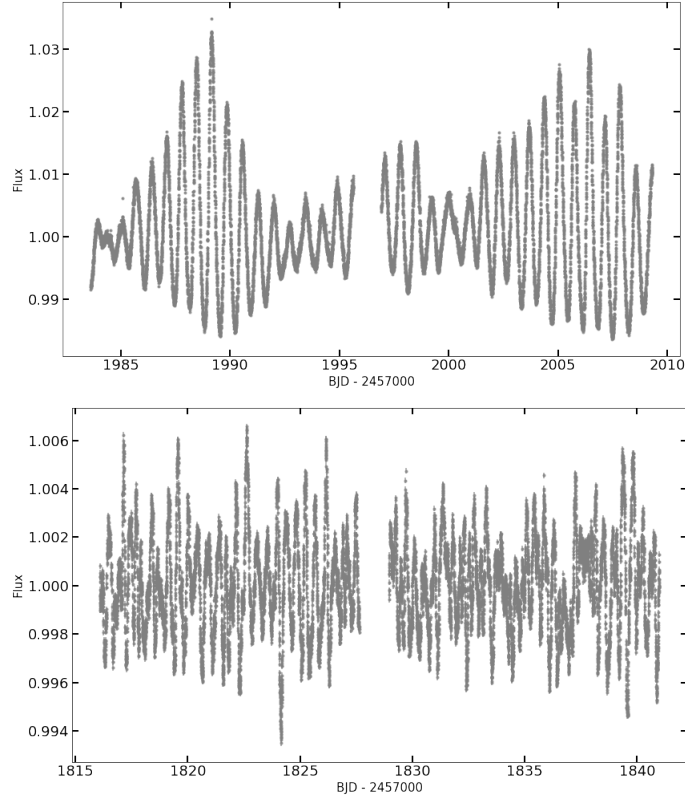


Fig. 6.5.— Several of our target stars exhibit photometric variability that is inconsistent with rotation. **Top:** A light curve of HIP 59767, a γ Doradus variable (?), shows a beat pattern making it difficult to measure an underlying rotation period. **Bottom:** A light curve of HIP 22361 shows stochastic variability that is not attributed to rotation.

are therefore not candidates for deriving gyrochronological ages. HIP 41274 is within the `gyro-interp` temperature range, but it is a RS-CVn variable (?), so it is also not a candidate for deriving a gyrochronological age. The close binary to HIP 41274 could be the source of its observed acceleration.

7. Combined Age Posteriors

For stars with $B - V \leq 0.58$ (corresponding to an F9.5 spectral type and approximate main-sequence mass of $1.1M_{\odot}$), we apply additional constraints on their ages based on stellar lifetimes and their CMD positions (e.g. ??). Using this technique, we also produce age posteriors for stars too massive for Li, R'_{HK} , or gyrochronology. CMD age posteriors are especially useful in cases where we can only derive a lower limit on the age from a lithium equivalent width upper limit because the stellar lifetime can be used to constrain the upper bound on the age (e.g. late F stars). To calculate CMD position-based age posteriors, we follow the method from ?, which uses a Bayesian approach to compute ages from stellar models and optical photometry. For each star, MCMC is used to explore the 5-dimensional parameter space in stellar mass, age, metallicity, initial rotation, inclination of the rotation axis, and parallax. We use the same priors on these parameters as in ?, and a Gaussian likelihood for the *Gaia* G_B , G , and G_R catalog photometry and errors. The model predictions at each MCMC step are

generated, as in ?, from a combination of MESA (?) evolutionary models with ATLAS9 model atmospheres (?). As in ?, for stars with $B - V \leq 0.27$, we include a prior based on the metallicity distribution of stars < 1 Gyr from ?.

For 55 stars more than one age tracer is available, and we improve our constraints by taking the product of each age posterior for a given star. In Table ?? we present our full 166-star sample, including basic properties and final median ages and 68% and 95% confidence intervals for each star. Figure ?? shows the age posteriors for the sample as a function of $B - V$.

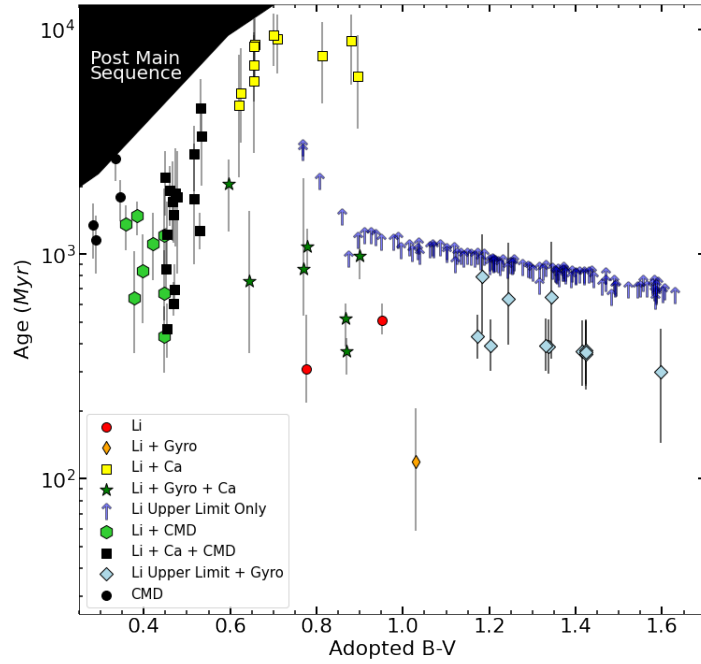


Fig. 7.1.— The medians and 68% confidence intervals for our derived age posteriors for the full sample of 166 stars, as a function of $B - V$. The symbol and color of each star represent which age tracer (or tracers) was used to derive the age posterior. The error bars represent the 68% confidence intervals.

Table 7.1. Our complete sample of accelerating stars with age posteriors and age tracers used. Ages come from a combination of lithium (Li), lithium upper limits (Li UL), calcium emission (R’HK), rotation periods (Gyro), and CMD positions (CMD). Age posteriors are either presented as a median with 68% and 95% confidence intervals or as 68%, 95% and 99.7% lower limits.

Target	R.A.	Dec.	G mag	Adopted $B - V$	Simbad Spec. Type	Dist (pc)	Tracers	Median Age or 68% LL (Myr)	68% CI or 95% LL (Myr)
HIP 143	00 01 49.35	+21 27 45.01	9.9811 ± 0.0002	1.033 ± 0.019	K3	43.4 ± 0.59	Li UL	≥ 4442.08	≥ 1034.76
HIP 669	00 08 16.36	-14 49 28.16	6.8956 ± 0.0002	0.624 ± 0.015	G1V	27.31 ± 0.02	Li, R’HK	5234.81	3132.71-8290.74
HIP 1068	00 13 15.83	+69 19 36.76	12.0476 ± 0.0015	1.315 ± 0.1	M4	20.52 ± 0.05	Li UL	≥ 4217.04	≥ 720.14
HIP 1412	00 17 40.90	-08 40 56.14	10.2711 ± 0.0006	1.414 ± 0.005	K7V	32.05 ± 0.09	Li UL, Gyro	367.99	258.82-508.19
HIP 1539	00 19 12.39	-03 03 13.01	10.2808 ± 0.0004	1.375 ± 0.005	K5V	31.19 ± 0.02	Li UL	≥ 4294.84	≥ 828.91
HIP 1771	00 22 25.19	+06 40 03.16	10.6176 ± 0.0005	1.489 ± 0.1	K7V	43.88 ± 0.09	Li UL	≥ 4234.75	≥ 744.83
HIP 2426	00 30 55.58	+73 22 14.89	10.0247 ± 0.0002	1.359 ± 0.02	K	29.48 ± 0.01	Li UL	≥ 4298.94	≥ 834.64
HIP 2843	00 36 01.85	-05 34 14.58	6.6114 ± 0.0002	0.466 ± 0.007	F6V	44.84 ± 0.06	Li UL, R’HK, CMD	1715.26	1150.23-2585.26
HIP 3008	00 38 15.29	+52 19 55.52	9.8115 ± 0.0003	1.424 ± 0.02	M0.0V	23.02 ± 0.01	Li UL, Gyro	360.24	250.15-502.94
HIP 3293	00 42 01.42	+40 41 17.71	6.9164 ± 0.0003	0.452 ± 0.009	F5	49.58 ± 0.07	Li, R’HK, CMD	862.22	651.1-1311.35
HIP 3633	00 46 34.04	+14 19 31.61	9.6005 ± 0.0001	0.814 ± 0.061	K3V	46.51 ± 0.21	Li UL, R’HK	7664.88	4696.93-10795.26
HIP 3724	00 47 48.47	+06 01 54.92	10.963 ± 0.0004	1.4 ± 0.02	M0.0V	40.62 ± 0.05	Li UL	≥ 4277.49	≥ 804.65
HIP 4024	00 51 34.56	-14 53 26.46	9.4072 ± 0.0002	1.069 ± 0.009	K3.5IV-V	43.46 ± 0.15	Li UL	≥ 4471.25	≥ 1075.56
HIP 5319	01 08 01.34	+32 00 43.66	6.1346 ± 0.0003	0.398 ± 0.01	F5IV	42.9 ± 0.06	Li UL, CMD	843.22	495.26-1230.45
HIP 5763	01 13 58.86	+16 29 40.26	9.3939 ± 0.0002	1.22 ± 0.015	K6V	32.01 ± 0.03	Li UL	≥ 4348.13	≥ 903.42
HIP 6776	01 27 06.32	+34 22 39.18	6.1623 ± 0.0003	0.453 ± 0.007	F7IVsv	42.05 ± 0.05	Li UL, R’HK, CMD	1220.64	830.65-1719.17
HIP 6796	01 27 27.60	+23 14 40.42	9.4186 ± 0.0004	1.157 ± 0.074	K7V	37.17 ± 0.04	Li UL	≥ 4400.69	≥ 976.89
HIP 6833	01 27 55.56	-01 59 29.35	8.4502 ± 0.0002	0.656 ± 0.002	G5V	48.1 ± 0.06	Li, R’HK	6929.95	4767.61-9685.66
HIP 6943	01 29 26.45	+42 16 54.85	7.078 ± 0.0003	0.453 ± 0.011	F5	48.19 ± 0.06	Li, R’HK, CMD	466.38	344.95-600.94
HIP 7287	01 33 53.62	+27 14 08.74	10.5178 ± 0.0002	1.2 ± 0.02	K5	48.57 ± 0.22	Li UL	≥ 4334.42	≥ 884.25
HIP 8653	01 51 31.19	-07 44 23.57	8.4607 ± 0.0003	0.644 ± 0.019	G5V	47.89 ± 1.09	Li, Gyro, R’HK	759.94	364.45-1561.09
HIP 9067	01 56 43.73	+29 48 39.04	10.8658 ± 0.0003	1.47 ± 0.1	K5V	47.3 ± 0.03	Li UL	≥ 4275.7	≥ 801.85
HIP 9989	02 08 40.08	-00 21 35.61	10.4039 ± 0.0003	1.248 ± 0.002	K5V	39.45 ± 0.03	Li UL	≥ 4340.02	≥ 892.08
HIP 10050	02 09 23.11	+17 13 27.11	6.2937 ± 0.0003	0.471 ± 0.001	F5V	36.32 ± 0.04	Li, R’HK, CMD	694.63	612.9-897.62
HIP 10532	02 15 41.72	-09 00 47.23	8.877 ± 0.0002	0.88 ± 0.033	K0V	44.19 ± 0.03	Li UL, R’HK	8922.3	5685.95-11625.28
HIP 11090	02 22 50.30	+41 23 46.65	5.7433 ± 0.0009	0.289 ± 0.006	F0III-IV	48.33 ± 0.14	CMD	1155.1	820.23-1479.52
HIP 11815	02 32 23.18	+03 22 57.48	11.2101 ± 0.0014	1.357 ± 0.02	K5	49.26 ± 0.05	Li UL	≥ 4276.65	≥ 803.47
HIP 12837	02 45 01.09	-22 09 59.51	7.0178 ± 0.0002	0.517 ± 0.008	F7V	38.9 ± 0.03	Li, R’HK, CMD	1771.7	899.96-3111.02
HIP 13681	02 56 14.10	-11 50 49.84	9.8618 ± 0.0003	1.062 ± 0.015	–	47.51 ± 0.04	Li UL	≥ 4456.4	≥ 1054.8

$=0_{\text{pt}}$

7.1. Stars within the Lithium Dip

There are several stars in our sample that are near the lithium dip, which is a range of $B - V$ ($\sim 0.4 - 0.55$) where lithium depletes rapidly around the age of the Hyades. This rapid evolution means the feature is not well modeled (e.g. ?), we discuss the consistency of the ages for stars near the lithium dip below.

Many stars with $0.4 < B - V < 0.55$ have age posteriors from different tracers that are generally consistent, including the lithium posterior. For example, HIP 42343 has ages from lithium (4964^{+4688}_{-2766} Myr), R'_{HK} (4793^{+3345}_{-2103} Myr), and CMD position (2105^{+1309}_{-1071} Myr) that give a combined posterior of 2791^{+933}_{-895} Myr. Similarly, the posteriors mainly agree for HIP 17118, HIP 12837, HIP 71899, and HIP 27021.

HIP 47013 shows significant offset between posteriors. The lithium non-detection gives a 2σ lower limit of 1528 Myr, which is consistent with the R'_{HK} posterior of 4997^{+3340}_{-2175} Myr. However, the CMD posterior gives a significantly younger age of 1181^{+247}_{-203} Myr. This may be due to the small number of stars within the lithium dip in the $\sim 500 - 1000$ Myr clusters used to calibrate **BAFFLES**. Because of this limitation, we recommend not relying on ages from lithium alone for stars within the lithium dip. We see similar discrepancies between the posteriors for HIP 6943, HIP 113159, HIP 10050, HIP 71243, HIP 3293, HIP 6776, HIP 44162, HIP 2843, HIP 115280, and HIP 21066.

8. Discussion

From our full sample of 166 stars, 102 age posteriors are based on lithium upper limits alone, while 55 stars have an age posterior based on more than one tracer. 31 of our targets ($\sim 19\%$) have a median age younger than 1 Gyr, which is more than expected for a uniform star formation rate. We discuss the full age distribution of our sample below.

8.1. Star Formation History of the Solar Neighborhood

? and ? found that the local star formation rate has been near constant for the last 10 to 12 Gyr based on the luminosity distribution of white dwarfs. Other studies found slightly different histories: a maximum star formation rate (~ 2 times the typical star formation rate at other times) ~ 3 Gyr ago (?) or two star forming events separated by ~ 5 Gyr (?). We assume a uniform star formation history, which also matches the priors used in BAFFLES (?) and gyro-interp (?).

8.2. Population Level Ages

We investigate how significant it is that our sample contains more stars than expected with median ages younger than 1 Gyr. Given the small size of our sample (< 200 stars), and the biased nature in which it was selected, we are not attempt-

ing here to constrain the local star formation history. Rather, we are examining whether the total age distribution we observe is broadly consistent with a uniform star formation rate in the solar neighborhood. We construct a population-level age distribution by summing (and then normalizing) the age posteriors for the entire sample. This is equivalent to marginalizing over individual stars; in the limit where each age posterior is a delta-function, this method produces a histogram of stellar ages. With a large enough volume-limited sample, we would expect this population-level distribution to match the star formation of the solar neighborhood. If the star formation rate has been uniform from 0 to 13 Gyr, we expect 1/13 (7.7%) of the summed age distribution to be younger than 1 Gyr.

The left panel of Figure ?? shows the population-level age distribution for our 166-star sample, which peaks at young ages ($\lesssim 2$ Gyr). Part of this peak is due to early-type stars, which have short main sequence lifetimes and thus do not trace the full 13 Gyr star formation history. These stars make up $\sim 19\%$ of the sample which is an over-representation relative to the initial mass function (???). This is likely due to the multiplicity fractions ($\sim 50 - 70\%$) of these stars (?) making them more likely to be accelerating. Removing stars with $B - V < 0.595$ (corresponding to a main sequence mass of $1.06 M_{\odot}$) leaves us with 135 stars and produces the posterior in the right panel of Figure ?. The peak at young ages is still present, but less pronounced.

This general behavior is consistent with the types of age posteriors we produce here. For any given late-type star the posterior is either a well-constrained Gaussian-like peak or a lower age limit (corresponding to a lithium upper limit) with only the youngest ages ruled out. While R'_{HK} produces Gaussian-like posteriors at both young and old ages for stars earlier than K2, lithium and gyrochronology generally produce peaked posteriors for later spectral types if they are younger than ~ 1 Gyr. Stars older than ~ 1 Gyr mostly do not have detectable lithium (depending on spectral type), and either have rotation periods longer than a typical *TESS* sector or are less spotted. For the reduced sample of 135 stars, we expect ~ 10 (7.7%) stars to be younger than 1 Gyr. Following the Poisson distribution, we calculate a 65% chance of between 8 and 13 stars being younger than 1 Gyr and a 96% chance of between 5 and 17 stars being younger than 1 Gyr. Testing this prediction is complicated by the fact that we have somewhat broad age posteriors, rather than delta-function age measurements. Nevertheless, we can estimate the number of young stars in our sample by summing the probability in the posteriors for all stars. We calculate that 18% of the age posterior for the reduced sample falls below 1 Gyr. This corresponds to ~ 24 stars, which is significantly higher than expected. In fact, for a Poisson distribution with an expectation value of 10.4 stars, we only expect a value of 24 or higher 0.01% of the time.

8.3. Testing Individual Age Tracers

Figure ?? further divides the late spectral type stars. The left panel only includes posteriors from lithium detections and/or rotation period detections. The right panel only includes posteriors derived from lithium upper limits. The posterior derived exclusively from lithium upper limits, as expected, is much less peaked and is essentially following the prior. The posterior derived from lithium detections and rotation rates peaks at ages younger than ~ 1 Gyr. This peaked posterior is expected for age tracers that only return peaked posteriors for the youngest stars.

Nevertheless, this peak seems to include more probability at ages less than 1 Gyr than we would expect from a uniform star formation rate operating between 0 and 13 Gyr. We investigate the relative contributions of lithium and gyrochronology to this result in Figure ?. We consider ages from lithium, calcium, and gyrochronology separately.

Lithium: We find that the combined age posterior for lithium-only derived posteriors (114 posteriors) is consistent with a uniform star formation rate. From the Poisson distribution and an assumed uniform star formation rate history, there is an 82% chance of between $5 \leq N_{stars} \leq 11$ being younger than 1 Gyr. We calculate 8.0% of the posterior (~ 9 stars) falls below 1 Gyr, consistent with this prediction.

Calcium: We derive age posteriors from R'_{HK} for 18 late-type stars. There are 7 additional late-type stars that have appropriate $B - V$ values to derive posteriors from R'_{HK} but have $R'_{HK} \leq -5$, which is outside the calibration range of **BAFFLES**, but consistent with old ages. From the Poisson distribution, there is a $\sim 60\%$ chance of measuring between 0 and 1 stars younger than 1 Gyr from a sample of 18 stars. 1% of our 18 star R'_{HK} posterior falls below 1 Gyr, which corresponds to finding 0.18 stars being less than 1 Gyr, which is again consistent with the prediction.

Gyrochronology: We have *TESS* light curves for 129 stars, 106 of which have $B - V > 0.595$. We measure rotation periods for 20 of the late-type stars. From the gyrochronology population-level age distribution, 92% of the probability is less than 1 Gyr, which corresponds to $\sim 17\%$ of the total late-type gyrochronology sample of 106 stars, or about 18 young stars. Our ages from gyrochronology do not include upper limits because flat *TESS* light curves could indicate a long rotation period (longer than a typical *TESS* sector) or just a star without significant star spots. Overall, we would expect ~ 8 stars out of 106 to have an age of 1 Gyr or younger. From the Poisson distribution with an expectation value of 8, there is a $\sim 4\%$ chance of finding 13 or more stars below 1 Gyr, and a $\sim 0.08\%$ chance of finding 18 or more. Thus, while the number of young stars identified from lithium and calcium alone are consistent with the predictions of a uniform star formation rate, there appears to be $\sim 5 - 10$ excess young stars identified by

rotation.

This result, that our combined age posterior suggests 18.0 stars are younger than 1 Gyr, does not change significantly even if we increase the size of the rotational period errors when computing age posteriors. Our period errors are between 0.11 days and 2.02 days, with a median of 0.55 days. Setting a minimum error bar to 1 day, 2 days, or 3 days still results in an estimate that 18.4, 17.8, and 16.8 stars are younger than 1 Gyr, respectively. As a result, we conclude that this excess of probability at young ages cannot be attributed solely to a systematic underestimate of the rotation period uncertainties.

There are several possible explanations for the higher-than-expected fraction of young stars in the gyrochronological sample. Since we select stars that are accelerating, there is a greater chance of selecting a short-period stellar binary (e.g. ?). The rotational evolution of stars in close binaries could be influenced by tides resulting in shorter rotation periods than expected from age alone. Additionally, *TESS* pixels may be contaminated by multiple stars (either physical binaries or crowded fields). While we screen for contaminated pixels, additional components may be present in some light curves. Alternatively, this effect could be physical: if binaries become unbound over time, a sample of accelerating stars (which is more likely to contain binaries than the field) would be younger than a volume-limited sample.

While we screened our initial accelerating star sample for binaries from WDS, SB9, and archival imaging, such a search is not complete to all binary configurations. Indeed, by constructing our sample from accelerating stars not in these catalogs we expect undiscovered binaries to be overrepresented in our sample compared to an unbiased volume-limited sample. For example, three eclipsing binaries (HIP 16247, HIP 45731, and HIP 49577) discussed in Section ?? were discovered to be physical binaries only after our survey began. We are currently carrying out radial velocity monitoring of the full sample with the aim of identifying additional short-period stellar binaries, and will report the results in a future paper.

8.4. Companions

Since the beginning of this survey in 2020, two brown dwarf companions have been discovered around two of the accelerating stars discussed in this work. ? found a $\sim 28 M_{\text{Jup}}$ companion to HIP 21152, a likely member of the Hyades Cluster (~ 750 Myr; ?). We measure a slightly older age 1116^{+406}_{-351} Myr based on a lithium upper limit and the star’s CMD position. This star, however, is in the lithium dip as discussed in section ?. The CMD age 855^{+1348}_{-412} , is consistent with the age of the Hyades. The lithium lower limit suggest an age a factor of two older than the Hyades (>1588 Myr), though this may be an overestimate due to how BAFFLES models the lithium dip.

? found a $\sim 31 M_{\text{Jup}}$ companion to HIP 5319. With $B - V < 0.45$, HIP 5319 is not a candidate for deriving an age from R'_{HK} using BAFFLES. ? report an age between $1.23^{+0.54}_{-0.6}$ Gyr by fitting isochrones from ??. We measure an age of 843^{+387}_{-348} Myr from a lithium upper limit and the star's CMD position, which is consistent with the previously reported age. These detections underscore how effective astrometric selection can be at identifying stars hosting substellar companions.

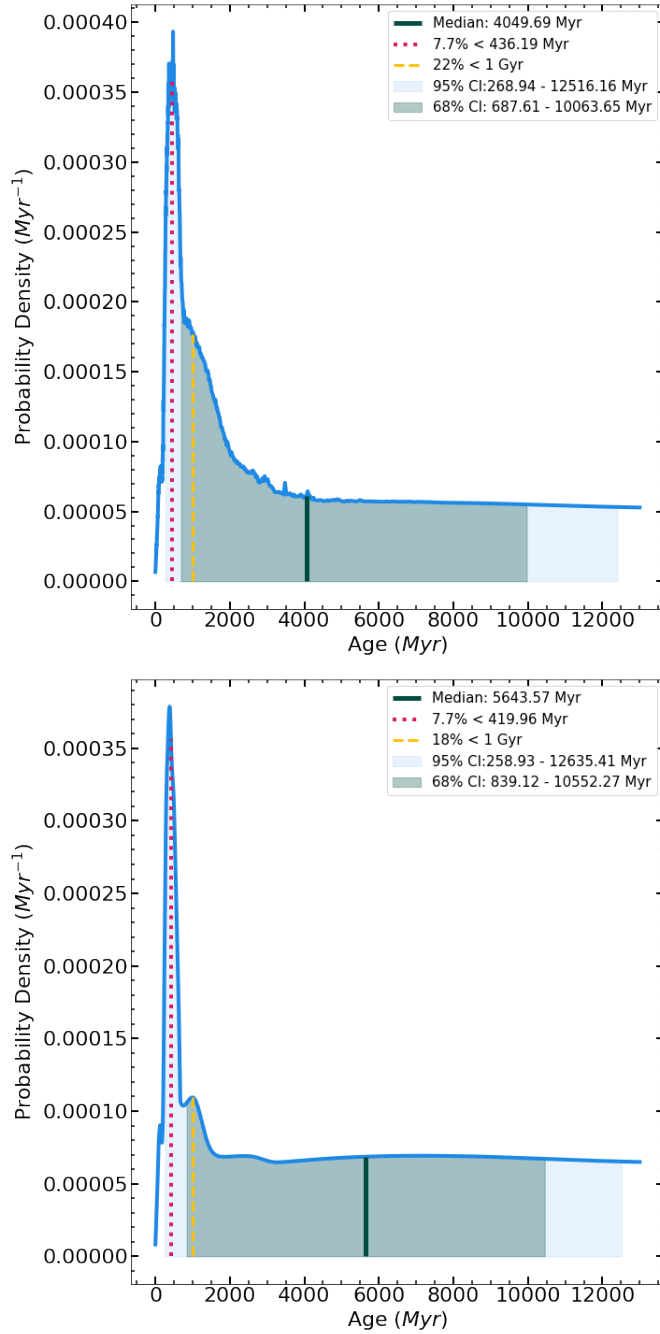


Fig. 8.1.— **Left:** Sum of all the age posteriors for the full sample. The distribution of ages is peaked at young ages, but this is partially due to the inclusion of early-type stars with shorter main sequence lifetimes. **Right:** Sum of age posteriors for only stars with $B - V > 0.595$, where there is still a peak at young ages, if less pronounced than for the full sample.

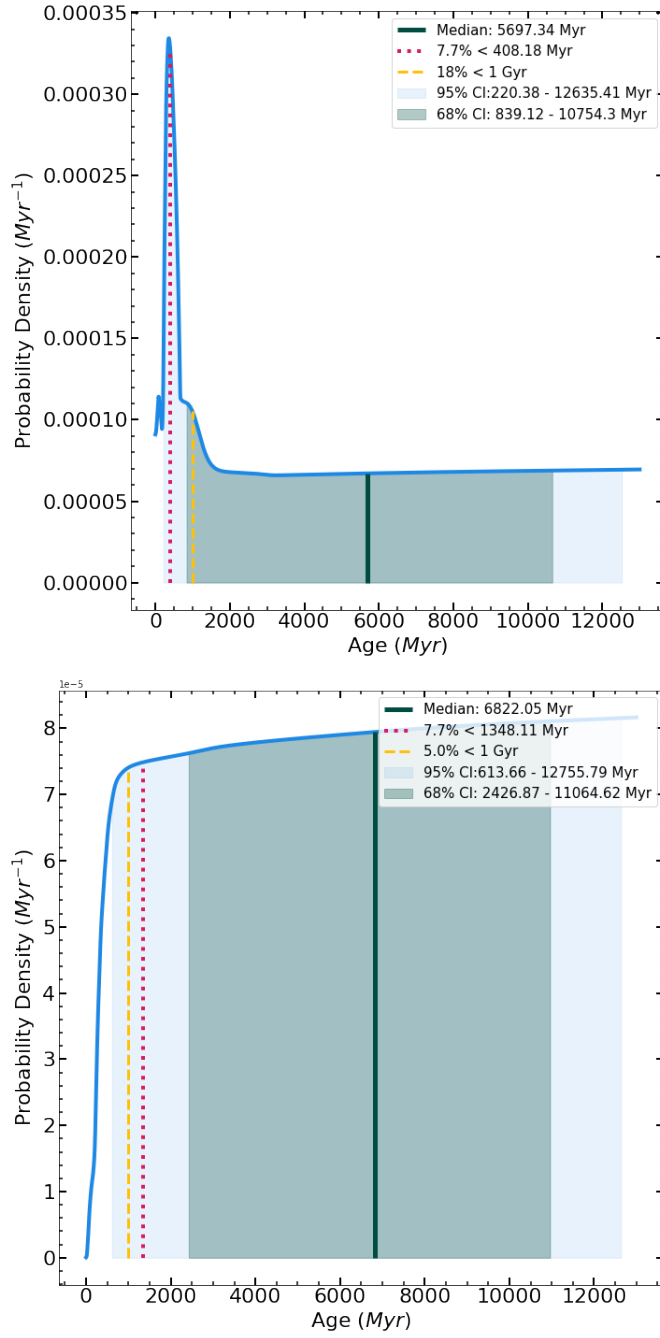


Fig. 8.2.— **Left:** The combined age posterior from lithium detections and/or gyrochronology for the late-type stars in the sample. The peak at young ages is consistent with stars that are < 1 Gyr being most likely to have detectable lithium or clear rotation signals. **Right:** For stars that only have lithium upper limits, the age posterior is mostly flat, following the prior, with only the youngest ages excluded.

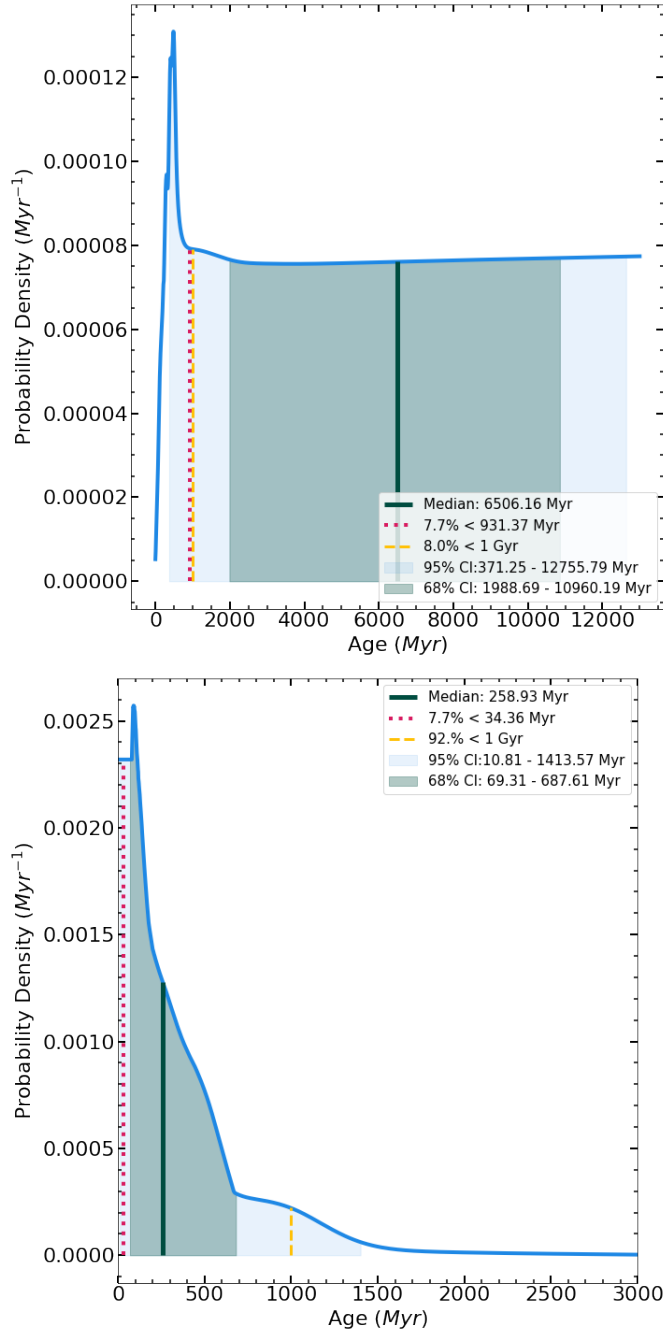


Fig. 8.3.— **Left:** Ages derived from lithium alone yield an age distribution consistent with a constant star formation rate. 8% of the combined distribution falls below 1 Gyr, which is consistent with the expected 7.7%. **Right:** Ages derived from gyrochronology alone yield a significantly younger age distribution than expected. For the 20 late-type stars with rotation periods $\lesssim 15$ days, 92% of the combined distribution of ages falls below 1 Gyr, which corresponds to $\sim 17\%$ of the full gyrochronology sample.

9. Conclusions

We identified a sample of 166 accelerating stars that may host substellar companions and do not have archival spectra. From APO/ARCES spectra of the full 166 star sample we measured lithium equivalent widths (or upper limits) and R'_{HK} . We acquired TESS light curves for 129 stars and measured rotation rates for 23 out of the 129 stars. For the earlier spectral types, we also constrain the ages from color-magnitude diagram positions. Combining these methods we calculated median ages and 68% and 95% confidence intervals for the full sample. We are continuing to collect APO/ARCES spectra of the full sample over at least 3 epochs per star to identify stellar binaries (which could be the source of the astrometric acceleration).

We find approximately twice as many young stars from gyrochronology than expected from a uniform star formation rate. We identify two possible explanations for this result. First, there could be unidentified short-period binaries in the sample: either short period binaries that spin each other up, or visual binaries that overlap in *TESS* images. Second, if binaries tend to dissolve over time, then young stars may be more likely to be part of binaries (and possess astrometric accelerations) than older stars. We expect our multi-epoch radial velocity monitoring to identify additional binaries in the sample, and we will report on the results of this monitoring in a future paper.

Looking ahead, *Gaia* DR4 is expected to be released in mid-2026. Unlike DR3, which only contains a single position and velocity measurement, DR4 will contain individual scans over several years. These individual scans will better differentiate between short-period stellar binaries and long-period substellar companions. Future *TESS* data releases will provide light curves for more accelerating stars, allowing for additional gyrochronological age measurements. Additional light curves for accelerating stars that already have light curves will allow for monitoring of spot evolution over the course of a stellar cycle.

The youngest of the accelerating stars in our sample are high-priority targets for direct imaging since these substellar companions will be bright enough to be imaged. In addition to continued radial velocity monitoring, we have begun a direct imaging campaign targeting the youngest stars. Results from both studies will be presented in future work. Substellar companions found in this sample will become key benchmarks for substellar evolution models.

Table 1.: We present our lithium equivalent width measurements or upper limits for a star and its corresponding **BAFFLES** median age and confidence intervals where appropriate. For stars with $B - V$ outside the **BAFFLES** range, we do not derive age priors.

Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)
HIP 143	1.033 ± 0.019		≤ 7.64	—	—		—	

≥ 484.61
 ≥ 1034

Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)		
HIP 669	0.624 ± 0.015		31.45 ± 1.89	4612.67	2164.81-9235.42	859.53-12375.61	–	–	–	–
HIP 1068	1.315 ± 0.1		≤ 68.37	–	–	–	≥ 138.08	≥ 720.		
HIP 1412	1.414 ± 0.005		≤ 13.65	–	–	–	≥ 254.02	≥ 803.		
HIP 1539	1.375 ± 0.005		≤ 12.17	–	–	–	≥ 280.46	≥ 828.		
HIP 1771	1.489 ± 0.1		≤ 15.03	–	–	–	≥ 164.0	≥ 744.		
HIP 2426	1.359 ± 0.02		≤ 10.93	–	–	–	≥ 285.77	≥ 834.		
HIP 2843	0.466 ± 0.007		≤ 7.97	–	–	–	≥ 716.16	≥ 1300		
HIP 3008	1.424 ± 0.02		≤ 14.98	–	–	–	≥ 246.95	≥ 797.		
HIP 3293	0.452 ± 0.009		13.22 ± 4.98	726.52	573.26-1064.21	224.61-1842.72	–	–	–	–
HIP 3633	0.814 ± 0.061		≤ 11.11	–	–	–	≥ 571.54	≥ 1376		
HIP 3724	1.4 ± 0.02		≤ 16.09	–	–	–	≥ 255.04	≥ 804.		
HIP 4024	1.069 ± 0.009		≤ 4.33	–	–	–	≥ 530.93	≥ 1075		
HIP 5319	0.398 ± 0.01		≤ 35.97	–	–	–	≥ 247.18	≥ 1301		
HIP 5763	1.22 ± 0.015		≤ 8.14	–	–	–	≥ 352.3	≥ 903.		
HIP 6776	0.453 ± 0.007		≤ 16.62	–	–	–	≥ 573.96	≥ 1158		
HIP 6796	1.157 ± 0.074		≤ 5.35	–	–	–	≥ 409.71	≥ 976.		

Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)		
HIP 6833	0.656 ± 0.002		13.12 ± 2.19	7124.14	4221.16-10604.78	1831.07-12609.77			–	–
HIP 6943	0.453 ± 0.011		45.34 ± 4.14	361.64	185.75-490.47	50.47-662.73			–	–
HIP 7287	1.2 ± 0.02		≤ 12.3	–	–	–			≥ 332.72	≥ 884.
HIP 8653	0.644 ± 0.019		47.98 ± 2.44	2101.63	818.22-7067.98	347.5-11896.89			–	–
HIP 9067	1.47 ± 0.1		≤ 6.96	–	–	–			≥ 231.94	≥ 801.
HIP 9989	1.248 ± 0.002		≤ 7.92	–	–	–			≥ 341.68	≥ 892.
HIP 10050	0.471 ± 0.001		14.76 ± 2.12	641.25	536.76-745.32	166.15-976.21			–	–
HIP 10532	0.88 ± 0.033		≤ 4.38	–	–	–			≥ 647.91	≥ 1305
HIP 11090	0.289 ± 0.006		≤ 37.8	–	–	–			–	–
HIP 11815	1.357 ± 0.02		≤ 21.0	–	–	–			≥ 253.41	≥ 803.
HIP 12837	0.517 ± 0.008		82.5 ± 4.08	2947.74	458.78-8919.31	118.81-12361.88			–	–
HIP 13681	1.062 ± 0.015		≤ 5.91	–	–	–			≥ 509.53	≥ 1054
HIP 13782	0.238 ± 0.003		≤ 25.62	–	–	–			–	–
HIP 15211	0.936 ± 0.007		≤ 3.61	–	–	–			≥ 609.82	≥ 1187
HIP 16247	1.02 ± 0.037		≤ 9.35	–	–	–			≥ 461.81	≥ 1015
HIP 16709	1.225 ± 0.02		≤ 11.19	–	–	–			≥ 328.75	≥ 879.

Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)		
HIP 17118	0.478 ± 0.023		40.49 ± 10.51	36	16.84	512.59-9572.67	172.65-12482.35		—	—
HIP 17213	1.1 ± 0.015		≤ 6.23	—	—	—	—		≥ 453.14	≥ 1004
HIP 18527	0.899 ± 0.021		≤ 4.12	—	—	—	—		≥ 633.73	≥ 1222
HIP 19739	1.424 ± 0.02		≤ 11.89	—	—	—	—		≥ 257.67	≥ 807
HIP 19912	1.148 ± 0.002		≤ 4.65	—	—	—	—		≥ 433.87	≥ 987
HIP 20419	1.183 ± 0.005		≤ 13.39	—	—	—	—		≥ 334.12	≥ 885
HIP 20648	0.049 ± 0.007		≤ 3.04	—	—	—	—		—	—
HIP 21066	0.472 ± 0.013		≤ 31.26	—	—	—	—		≥ 391.93	≥ 1070
HIP 21152	0.42 ± 0.014		≤ 13.2	—	—	—	—		≥ 566.24	≥ 1588
HIP 22361	0.283 ± 0.004		≤ 26.75	—	—	—	—		—	—
HIP 23208	1.18 ± 0.006		≤ 7.62	—	—	—	—		≥ 376.02	≥ 927
HIP 24017	0.448 ± 0.004		≤ 9.48	—	—	—	—		≥ 669.48	≥ 1254
HIP 24177	1.461 ± 0.005		≤ 33.19	—	—	—	—		≥ 159.14	≥ 719
HIP 24292	1.23 ± 0.015		≤ 6.29	—	—	—	—		≥ 365.54	≥ 916
HIP 24457	1.118 ± 0.005		≤ 3.42	—	—	—	—		≥ 482.73	≥ 1038
HIP 25606	0.807 ± 0.027		≤ 3.03	—	—	—	—		≥ 1021.51	≥ 2111

Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)		
HIP 27021	0.531 ± 0.006		44.96 ± 6.8	6549.83	2786.96-10753.71	830.61-12664.69	–	–		
HIP 28823	0.358 ± 0.005		≤ 38.81	–	–	–	≥ 199.04	≥ 1161		
HIP 29208	0.895 ± 0.022		≤ 6.74	–	–	–	≥ 568.66	≥ 1152		
HIP 33282	1.344 ± 0.035		≤ 13.59	–	–	–	≥ 278.21	≥ 827		
HIP 33368	1.346 ± 0.1		≤ 8.69	–	–	–	≥ 288.0	≥ 847		
HIP 34804	1.591 ± 0.082		≤ 8.55	–	–	–	≥ 157.52	≥ 730		
HIP 36081	0.62 ± 0.005		49.7 ± 1.89	2660.91	1006.18-8005.41	403.71-12139.12	–	–		
HIP 37267	1.277 ± 0.022		≤ 8.48	–	–	–	≥ 325.41	≥ 875		
HIP 40518	1.134 ± 0.004		≤ 5.26	–	–	–	≥ 430.98	≥ 985		
HIP 41274	0.952 ± 0.011		7.47 ± 1.22	505.8	439.44-602.43	335.92-820.22	–	–		
HIP 41277	1.03 ± 0.015		125.14 ± 5.01	173.84	96.87-252.49	31.29-327.98	–	–		
HIP 41319	0.448 ± 0.005		29.11 ± 2.49	489.33	249.94-579.52	68.41-782.82	–	–		
HIP 42231	1.241 ± 0.015		≤ 16.83	–	–	–	≥ 296.87	≥ 847		
HIP 42343	0.516 ± 0.001		35.12 ± 2.7	4964.37	2198.09-9633.86	776.39-12466.6	–	–		
HIP 42507	1.361 ± 0.001		≤ 30.52	–	–	–	≥ 226.84	≥ 778		
HIP 43745	1.423 ± 0.02		≤ 12.37	–	–	–	≥ 256.35	≥ 806		

Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)		
HIP 44162	0.461 ± 0.005		≤ 6.86		—	—	—		≥ 735.01	≥ 1310
HIP 44953	0.767 ± 0.002		≤ 3.52		—	—	—		≥ 1323.58	≥ 2846
HIP 45621	0.869 ± 0.013		28.92 ± 4.34	409.4		336.01-508.61		157.11-770.0	—	—
HIP 45731	1.586 ± 0.02		≤ 23.11		—	—	—		≥ 91.63	≥ 650.
HIP 46385	1.586 ± 0.1		≤ 30.6		—	—	—		≥ 92.44	≥ 661
HIP 47013	0.53 ± 0.003		≤ 9.22		—	—	—		≥ 215.33	≥ 1527
HIP 47261	1.21 ± 0.015		≤ 8.15		—	—	—		≥ 356.79	≥ 908.
HIP 47300	0.223 ± 0.004		≤ 59.79		—	—	—		—	—
HIP 47387	1.07 ± 0.047		≤ 5.05		—	—	—		≥ 489.6	≥ 1046
HIP 47539	1.353 ± 0.011		≤ 15.01		—	—	—		≥ 270.07	≥ 819.
HIP 47741	1.542 ± 0.039		≤ 15.13		—	—	—		≥ 130.85	≥ 698.
HIP 48165	0.925 ± 0.015		≤ 3.19		—	—	—		≥ 638.59	≥ 1219
HIP 48629	1.581 ± 0.197		≤ 9.94		—	—	—		≥ 152.31	≥ 744.
HIP 49046	1.372 ± 0.015		≤ 12.76		—	—	—		≥ 276.52	≥ 825.
HIP 49526	1.492 ± 0.02		≤ 24.48		—	—	—		≥ 147.69	≥ 710.
HIP 49577	1.331 ± 0.014		≤ 6.65		—	—	—		≥ 318.63	≥ 867

Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)		
HIP 49910	1.391 ± 0.023		≤ 12.9	—	—	—	—	—	≥ 268.72	≥ 817.3
HIP 51208	1.2 ± 0.015		≤ 7.14	—	—	—	—	—	≥ 371.01	≥ 922.3
HIP 52339	1.42 ± 0.015		≤ 11.84	—	—	—	—	—	≥ 260.19	≥ 809.3
HIP 53175	1.293 ± 0.005		≤ 11.01	—	—	—	—	—	≥ 304.58	≥ 854.3
HIP 53869	0.868 ± 0.045		≤ 5.27	—	—	—	—	—	≥ 628.48	≥ 1339.3
HIP 54094	1.1 ± 0.02		≤ 4.44	—	—	—	—	—	≥ 481.2	≥ 1033.3
HIP 54199	0.874 ± 0.016		≤ 34.44	—	—	—	—	—	≥ 383.88	≥ 952.3
HIP 55192	0.776 ± 0.015		75.4 ± 1.89	308.96	218.51-539.13	66.28-1459.69	—	—	—	—
HIP 55409	0.658 ± 0.003		≤ 5.57	—	—	—	—	—	≥ 892.89	≥ 3214.3
HIP 57361	1.482 ± 0.015		≤ 70.46	—	—	—	—	—	≥ 85.43	≥ 646.3
HIP 57571	1.369 ± 0.02		≤ 11.45	—	—	—	—	—	≥ 281.31	≥ 830.3
HIP 57645	0.895 ± 0.015		≤ 3.69	—	—	—	—	—	≥ 659.12	≥ 1242.3
HIP 57984	1.316 ± 0.015		≤ 10.03	—	—	—	—	—	≥ 302.07	≥ 851.3
HIP 59000	1.336 ± 0.014		≤ 10.5	—	—	—	—	—	≥ 292.85	≥ 842.3
HIP 59198	1.409 ± 0.015		≤ 11.98	—	—	—	—	—	≥ 264.26	≥ 813.3
HIP 59199	0.334 ± 0.015		47.3 ± 3.74	—	—	—	—	—	—	—

Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)		
HIP 59247	1.29 ± 0.015		≤ 10.15	—	—	—	—	—	≥ 310.2	≥ 860.
HIP 59767	0.346 ± 0.011		32.5 ± 3.95	—	—	—	—	—	—	—
HIP 59905	1.125 ± 0.015		≤ 5.68	—	—	—	—	—	≥ 430.87	≥ 985.
HIP 59953	1.602 ± 0.1		≤ 25.28	—	—	—	—	—	≥ 96.84	≥ 665.
HIP 60829	0.71 ± 0.012		≤ 3.99	—	—	—	—	—	≥ 1515.85	≥ 3828
HIP 61947	1.582 ± 0.082		≤ 21.91	—	—	—	—	—	≥ 104.87	≥ 673.
HIP 62325	0.989 ± 0.007		≤ 1.63	—	—	—	—	—	≥ 585.8	≥ 1156
HIP 62627	0.995 ± 0.003		≤ 7.1	—	—	—	—	—	≥ 480.59	≥ 1042
HIP 63419	0.779 ± 0.021		7.65 ± 2.67	2340.25	1276.27-5485.81	757.54-10722.15	—	—	—	—
HIP 63421	1.354 ± 0.01		≤ 16.29	—	—	—	—	—	≥ 265.57	≥ 815.
HIP 63661	1.468 ± 0.1		≤ 17.8	—	—	—	—	—	≥ 166.58	≥ 747.
HIP 65120	1.49 ± 0.02		≤ 26.81	—	—	—	—	—	≥ 142.13	≥ 706.
HIP 65602	0.911 ± 0.002		≤ 3.47	—	—	—	—	—	≥ 641.92	≥ 1219
HIP 65651	1.203 ± 0.02		≤ 17.22	—	—	—	—	—	≥ 309.3	≥ 860.
HIP 65706	1.577 ± 0.082		≤ 10.88	—	—	—	—	—	≥ 144.27	≥ 717.
HIP 66262	1.165 ± 0.067		≤ 6.35	—	—	—	—	—	≥ 393.3	≥ 956.

Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)		
HIP 66315	1.039 ± 0.121		≤ 12.91	—	—	—	—	—	≥ 395.09	≥ 985.09
HIP 66587	1.419 ± 0.009		≤ 15.03	—	—	—	—	—	≥ 249.51	≥ 799.51
HIP 66828	1.33 ± 0.02		≤ 8.89	—	—	—	—	—	≥ 303.69	≥ 852.69
HIP 69311	1.172 ± 0.07		≤ 9.39	—	—	—	—	—	≥ 360.75	≥ 923.75
HIP 69333	1.173 ± 0.004		≤ 4.28	—	—	—	—	—	≥ 426.36	≥ 979.36
HIP 69860	1.629 ± 0.1		≤ 25.8	—	—	—	—	—	≥ 92.18	≥ 656.18
HIP 69963	1.39 ± 0.02		≤ 34.7	—	—	—	—	—	≥ 217.68	≥ 769.68
HIP 70472	1.257 ± 0.01		≤ 5.02	—	—	—	—	—	≥ 368.06	≥ 918.06
HIP 71243	0.45 ± 0.003		≤ 12.35	—	—	—	—	—	≥ 624.45	≥ 1202.45
HIP 71515	0.95 ± 0.002		≤ 4.52	—	—	—	—	—	≥ 566.76	≥ 1138.76
HIP 71899	0.533 ± 0.01		90.22 ± 3.36	3446.91	535.05-9289.75	124.97-12428.99	—	—	—	—
HIP 71957	0.385 ± 0.006		62.96 ± 6.43	742.99	198.67-2858.42	57.32-9369.38	—	—	—	—
HIP 73121	1.035 ± 0.009		≤ 3.93	—	—	—	—	—	≥ 544.57	≥ 1093.57
HIP 73183	1.555 ± 0.197		≤ 10.92	—	—	—	—	—	≥ 152.29	≥ 747.29
HIP 73252	1.565 ± 0.082		≤ 14.68	—	—	—	—	—	≥ 129.34	≥ 703.34
HIP 73787	1.21 ± 0.008		≤ 7.17	—	—	—	—	—	≥ 366.04	≥ 917.04

Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)		
HIP 74926	1.214 ± 0.009		≤ 8.04	—	—	—	—	—	≥ 356.02	≥ 907.3
HIP 79203	0.447 ± 0.007		52.15 ± 2.24	316.16	170.59-447.18	48.25-603.54	—	—	—	—
HIP 79593	1.584 ± 0.01		≤ 64.58	—	—	—	—	—	≥ 62.8	≥ 619.3
HIP 81655	1.605 ± 0.1		≤ 16.7	—	—	—	—	—	≥ 114.09	≥ 684.3
HIP 82260	0.767 ± 0.002		≤ 2.66	—	—	—	—	—	≥ 1383.07	≥ 2980.3
HIP 83676	1.015 ± 0.015		≤ 5.23	—	—	—	—	—	≥ 513.21	≥ 1071.3
HIP 83929	1.597 ± 0.197		≤ 7.89	—	—	—	—	—	≥ 164.45	≥ 754.3
HIP 87370	0.656 ± 0.015		5.45 ± 1.73	9440.7	5779.69-11965.31	2459.94-12855.22	—	—	—	—
HIP 88962	1.372 ± 0.004		≤ 7.43	—	—	—	—	—	≥ 302.95	≥ 850.3
HIP 89320	1.586 ± 0.197		≤ 17.74	—	—	—	—	—	≥ 119.63	≥ 709.3
HIP 89449	1.323 ± 0.019		≤ 13.44	—	—	—	—	—	≥ 284.08	≥ 833.3
HIP 90306	1.442 ± 0.02		≤ 20.2	—	—	—	—	—	≥ 220.21	≥ 771.3
HIP 93072	1.437 ± 0.004		≤ 9.38	—	—	—	—	—	≥ 266.15	≥ 815.3
HIP 98007	0.978 ± 0.003		≤ 4.06	—	—	—	—	—	≥ 561.44	≥ 1130.3
HIP 99969	1.14 ± 0.015		≤ 6.74	—	—	—	—	—	≥ 405.74	≥ 959.3
HIP 100133	1.43 ± 0.012		≤ 10.69	—	—	—	—	—	≥ 261.29	≥ 810.3

Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)		
HIP 101852	0.597 ± 0.012		≤ 5.55	—	—	—	—	—	≥ 413.67	≥ 2147
HIP 102119	1.121 ± 0.009		≤ 15.56	—	—	—	—	—	≥ 349.81	≥ 902
HIP 104687	0.654 ± 0.041		63.32 ± 3.17	1661.55	516.08-6885.97	216.02-11896.04	—	—	—	—
HIP 105504	1.255 ± 0.015		≤ 7.34	—	—	—	—	—	≥ 343.5	≥ 893
HIP 107317	1.522 ± 0.021		≤ 19.01	—	—	—	—	—	≥ 131.93	≥ 696
HIP 108036	0.378 ± 0.011		36.67 ± 8.24	1531.65	629.16-4486.11	186.82-10568.9	—	—	—	—
HIP 109807	1.252 ± 0.005		≤ 8.56	—	—	—	—	—	≥ 334.91	≥ 885
HIP 110106	1.399 ± 0.02		≤ 18.94	—	—	—	—	—	≥ 247.79	≥ 797
HIP 110401	1.208 ± 0.013		≤ 6.42	—	—	—	—	—	≥ 374.95	≥ 926
HIP 110663	0.77 ± 0.005		≤ 30.38	—	—	—	—	—	≥ 449.42	≥ 1135
HIP 111976	1.548 ± 0.1		≤ 17.49	—	—	—	—	—	≥ 126.14	≥ 705
HIP 113159	0.469 ± 0.005		22.57 ± 2.34	563.71	364.7-656.67	95.73-869.36	—	—	—	—
HIP 115280	0.468 ± 0.004		≤ 13.37	—	—	—	—	—	≥ 630.31	≥ 1208
HIP 115411	0.7 ± 0.005		≤ 3.79	—	—	—	—	—	≥ 1551.06	≥ 4259
HIP 116973	1.085 ± 0.008		≤ 3.36	—	—	—	—	—	≥ 525.19	≥ 1071
HIP 117159	0.859 ± 0.006		≤ 2.4	—	—	—	—	—	≥ 801.54	≥ 1475

Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)			
HIP 117235	1.376 ± 0.133		≤ 13.91	—	—	—	—	—	≥ 233.62	≥ 806.3	≥ 909.7
HIP 117410	1.244 ± 0.014		≤ 6.27	—	—	—	—	—	≥ 359.04	≥ 909.7	≥ 901.0
HIP 117795	1.21 ± 0.015		≤ 8.98	—	—	—	—	—	≥ 350.04	≥ 901.0	≥ 901.0
HIP 118086	1.3 ± 0.015		≤ 18.5	—	—	—	—	—	≥ 273.54	≥ 823.5	≥ 823.5
HIP 118310	1.187 ± 0.052		≤ 4.14	—	—	—	—	—	≥ 416.39	≥ 975.0	≥ 975.0

Table .2:: Each target star and its adopted $B - V$, measured S index and measured $\log(R'_{HK})$ are listed. R'_{HK} is only calibrated as an age metric for F5-K2 stars for which report a median age, 68% and 98% confidence intervals.

Target	Adopted B-V	S	$\log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)			
HIP 143	1.033 ± 0.019	0.41	-4.58	—	—	—	—	—	—
HIP 669	0.624 ± 0.015	0.11	-4.88	6468.35	3805.7-9673.65	1811.36-12362.31			
HIP 1068	1.315 ± 0.1	0.41	-4.3	—	—	—	—	—	—
HIP 1412	1.414 ± 0.005	1.38	-3.7	—	—	—	—	—	—
HIP 1539	1.375 ± 0.005	1.25	-3.77	—	—	—	—	—	—
HIP 1771	1.489 ± 0.1	1.4	-3.66	—	—	—	—	—	—

Target	Adopted B-V	S	$\log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 2426	1.359 ± 0.02	0.85	-3.95	—	—	—
HIP 2843	0.466 ± 0.007	0.1	-4.82	5038.2	2848.33-8376.32	1319.5-12018.59
HIP 3008	1.424 ± 0.02	1.57	-3.64	—	—	—
HIP 3293	0.452 ± 0.009	0.06	-4.8	4744.41	2658.9-8090.39	1223.88-11927.21
HIP 3633	0.814 ± 0.061	0.28	-4.93	7627.38	4629.95-10777.13	2245.46-12580.04
HIP 3724	1.4 ± 0.02	1.95	-3.56	—	—	—
HIP 4024	1.069 ± 0.009	0.39	-4.55	—	—	—
HIP 5319	0.398 ± 0.01	0.13	-4.83	—	—	—
HIP 5763	1.22 ± 0.015	0.67	-4.16	—	—	—
HIP 6776	0.453 ± 0.007	0.25	-4.89	6703.86	3969.45-9900.36	1896.83-12406.95
HIP 6796	1.157 ± 0.074	0.58	-4.28	—	—	—
HIP 6833	0.656 ± 0.002	0.12	-4.91	7056.53	4218.1-10241.09	2027.37-12469.75
HIP 6943	0.453 ± 0.011	0.06	-4.8	4719.62	2643.05-8065.82	1215.9-11919.12
HIP 7287	1.2 ± 0.02	0.44	-4.36	—	—	—
HIP 8653	0.644 ± 0.019	0.22	-4.91	7289.8	4385.03-10462.51	2115.46-12502.65
HIP 9067	1.47 ± 0.1	0.74	-3.94	—	—	—

Target	Adopted B-V	S	$\log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 9989	1.248 ± 0.002	0.78	-4.07	—	—	—
HIP 10050	0.471 ± 0.001	0.05	-4.8	4649.09	2597.95-7995.11	1193.23-11895.76
HIP 10532	0.88 ± 0.033	0.19	-5.0	8917.93	5673.64-11623.81	2817.71-12794.38
HIP 11090	0.289 ± 0.006	0.08	-4.82	—	—	—
HIP 11815	1.357 ± 0.02	2.34	-3.51	—	—	—
HIP 12837	0.517 ± 0.008	0.13	-4.84	5572.64	3199.05-8862.9	1498.1-12163.83
HIP 13681	1.062 ± 0.015	0.55	-4.42	—	—	—
HIP 13782	0.238 ± 0.003	0.11	-4.83	—	—	—
HIP 15211	0.936 ± 0.007	0.21	-4.95	—	—	—
HIP 16247	1.02 ± 0.037	1.24	-4.13	—	—	—
HIP 16709	1.225 ± 0.02	0.46	-4.32	—	—	—
HIP 17118	0.478 ± 0.023	0.14	-4.84	5506.82	3155.38-8805.1	1475.75-12147.31
HIP 17213	1.1 ± 0.015	0.8	-4.21	—	—	—
HIP 18527	0.899 ± 0.021	0.45	-4.72	3335.89	1782.91-6529.46	789.98-11322.22
HIP 19739	1.424 ± 0.02	1.01	-3.83	—	—	—
HIP 19912	1.148 ± 0.002	0.39	-4.47	—	—	—

Target	Adopted B-V	S	$\log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 20419	1.183 ± 0.005	0.47	-4.35	—	—	—
HIP 20648	0.049 ± 0.007	0.1	-4.76	—	—	—
HIP 21066	0.472 ± 0.013	0.13	-4.83	5415.69	3095.17-8724.59	1445.0-12123.82
HIP 21152	0.42 ± 0.014	0.12	-4.82	—	—	—
HIP 22361	0.283 ± 0.004	0.13	-4.83	—	—	—
HIP 23208	1.18 ± 0.006	1.03	-4.01	—	—	—
HIP 24017	0.448 ± 0.004	0.11	-4.82	—	—	—
HIP 24177	1.461 ± 0.005	0.65	-4.0	—	—	—
HIP 24292	1.23 ± 0.015	0.7	-4.13	—	—	—
HIP 24457	1.118 ± 0.005	0.45	-4.44	—	—	—
HIP 25606	0.807 ± 0.027	0.07	-5.1	—	—	—
HIP 27021	0.531 ± 0.006	0.05	-4.81	4955.97	2795.06-8297.47	1292.55-11993.99
HIP 28823	0.358 ± 0.005	0.11	-4.83	—	—	—
HIP 29208	0.895 ± 0.022	0.11	-5.13	—	—	—
HIP 33282	1.344 ± 0.035	1.12	-3.83	—	—	—
HIP 33368	1.346 ± 0.1	1.19	-3.81	—	—	—

Target	Adopted B-V	S	$\log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 34804	1.591 ± 0.082	0.38	-4.18	—	—	—
HIP 36081	0.62 ± 0.005	0.06	-4.86	6087.61	3544.88-9317.97	1676.01-12283.7
HIP 37267	1.277 ± 0.022	0.72	-4.08	—	—	—
HIP 40518	1.134 ± 0.004	0.48	-4.39	—	—	—
HIP 41274	0.952 ± 0.011	0.61	-4.53	—	—	—
HIP 41277	1.03 ± 0.015	1.68	-3.98	—	—	—
HIP 41319	0.448 ± 0.005	0.22	-4.88	—	—	—
HIP 42231	1.241 ± 0.015	0.53	-4.24	—	—	—
HIP 42343	0.516 ± 0.001	0.05	-4.8	4792.54	2689.78-8138.18	1239.43-11942.73
HIP 42507	1.361 ± 0.001	1.48	-3.7	—	—	—
HIP 43745	1.423 ± 0.02	1.28	-3.73	—	—	—
HIP 44162	0.461 ± 0.005	0.1	-4.82	5091.45	2882.9-8426.86	1337.04-12034.09
HIP 44953	0.767 ± 0.002	0.12	-5.01	—	—	—
HIP 45621	0.869 ± 0.013	0.36	-4.83	5312.27	3027.16-8631.79	1410.33-12096.23
HIP 45731	1.586 ± 0.02	4.53	-3.11	—	—	—
HIP 46385	1.586 ± 0.1	1.07	-3.73	—	—	—

Target	Adopted B-V	S	$\log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 47013	0.53 ± 0.003	0.06	-4.81	4997.36	2821.85-8337.18	1306.1-12006.47
HIP 47261	1.21 ± 0.015	1.16	-3.93	—	—	—
HIP 47300	0.223 ± 0.004	0.08	-4.83	—	—	—
HIP 47387	1.07 ± 0.047	0.49	-4.46	—	—	—
HIP 47539	1.353 ± 0.011	0.95	-3.9	—	—	—
HIP 47741	1.542 ± 0.039	0.61	-3.99	—	—	—
HIP 48165	0.925 ± 0.015	0.16	-5.03	—	—	—
HIP 48629	1.581 ± 0.197	1.11	-3.72	—	—	—
HIP 49046	1.372 ± 0.015	1.26	-3.77	—	—	—
HIP 49526	1.492 ± 0.02	0.97	-3.81	—	—	—
HIP 49577	1.331 ± 0.014	0.71	-4.05	—	—	—
HIP 49910	1.391 ± 0.023	1.09	-3.82	—	—	—
HIP 51208	1.2 ± 0.015	0.59	-4.23	—	—	—
HIP 52339	1.42 ± 0.015	1.32	-3.72	—	—	—
HIP 53175	1.293 ± 0.005	1.41	-3.77	—	—	—
HIP 53869	0.868 ± 0.045	0.36	-4.83	5401.81	3086.04-8712.37	1440.37-12120.17

Target	Adopted B-V	S	$\log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 54094	1.1 ± 0.02	0.16	-4.9	—	—	—
HIP 54199	0.874 ± 0.016	0.13	-5.09	—	—	—
HIP 55192	0.776 ± 0.015	0.1	-5.03	—	—	—
HIP 55409	0.658 ± 0.003	0.12	-4.91	7119.41	4262.9-10301.28	2050.96-12479.67
HIP 57361	1.482 ± 0.015	0.64	-4.0	—	—	—
HIP 57571	1.369 ± 0.02	0.73	-4.0	—	—	—
HIP 57645	0.895 ± 0.015	0.29	-4.87	6194.96	3617.97-9416.26	1713.82-12306.8
HIP 57984	1.316 ± 0.015	1.08	-3.87	—	—	—
HIP 59000	1.336 ± 0.014	1.96	-3.6	—	—	—
HIP 59198	1.409 ± 0.015	0.93	-3.87	—	—	—
HIP 59199	0.334 ± 0.015	0.1	-4.82	—	—	—
HIP 59247	1.29 ± 0.015	0.81	-4.01	—	—	—
HIP 59767	0.346 ± 0.011	0.12	-4.83	—	—	—
HIP 59905	1.125 ± 0.015	0.58	-4.32	—	—	—
HIP 59953	1.602 ± 0.1	0.97	-3.77	—	—	—
HIP 60829	0.71 ± 0.012	0.07	-4.96	8269.16	5128.5-11264.78	2514.96-12724.17

Target	Adopted B-V	S	$\log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 61947	1.582 ± 0.082	1.12	-3.71	—	—	—
HIP 62325	0.989 ± 0.007	0.08	-5.23	—	—	—
HIP 62627	0.995 ± 0.003	0.2	-4.91	—	—	—
HIP 63419	0.779 ± 0.021	0.36	-4.9	6919.61	4121.07-10109.29	1976.29-12446.08
HIP 63421	1.354 ± 0.01	0.93	-3.91	—	—	—
HIP 63661	1.468 ± 0.1	0.95	-3.84	—	—	—
HIP 65120	1.49 ± 0.02	0.99	-3.81	—	—	—
HIP 65602	0.911 ± 0.002	0.24	-4.92	—	—	—
HIP 65651	1.203 ± 0.02	1.16	-3.93	—	—	—
HIP 65706	1.577 ± 0.082	0.96	-3.78	—	—	—
HIP 66262	1.165 ± 0.067	0.48	-4.35	—	—	—
HIP 66315	1.039 ± 0.121	0.33	-4.66	—	—	—
HIP 66587	1.419 ± 0.009	0.81	-3.93	—	—	—
HIP 66828	1.33 ± 0.02	1.17	-3.83	—	—	—
HIP 69311	1.172 ± 0.07	1.15	-3.97	—	—	—
HIP 69333	1.173 ± 0.004	0.49	-4.34	—	—	—

Target	Adopted B-V	S	$\log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 69860	1.629 ± 0.1	1.22	-3.66	—	—	—
HIP 69963	1.39 ± 0.02	0.57	-4.1	—	—	—
HIP 70472	1.257 ± 0.01	0.65	-4.14	—	—	—
HIP 71243	0.45 ± 0.003	0.1	-4.81	5027.51	2841.38-8366.19	1315.99-12015.47
HIP 71515	0.95 ± 0.002	0.33	-4.77	—	—	—
HIP 71899	0.533 ± 0.01	0.16	-4.86	5942.96	3446.92-9187.67	1625.41-12251.95
HIP 71957	0.385 ± 0.006	0.11	-4.82	—	—	—
HIP 73121	1.035 ± 0.009	0.19	-4.89	—	—	—
HIP 73183	1.555 ± 0.197	1.13	-3.72	—	—	—
HIP 73252	1.565 ± 0.082	1.26	-3.67	—	—	—
HIP 73787	1.21 ± 0.008	0.76	-4.11	—	—	—
HIP 74926	1.214 ± 0.009	0.57	-4.23	—	—	—
HIP 79203	0.447 ± 0.007	0.07	-4.8	—	—	—
HIP 79593	1.584 ± 0.01	0.19	-4.48	—	—	—
HIP 81655	1.605 ± 0.1	0.6	-3.98	—	—	—
HIP 82260	0.767 ± 0.002	0.08	-5.03	—	—	—

Target	Adopted B-V	S	$\log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 83676	1.015 ± 0.015	0.37	-4.65	—	—	—
HIP 83929	1.597 ± 0.197	0.73	-3.9	—	—	—
HIP 87370	0.656 ± 0.015	0.07	-4.9	6908.91	4113.47-10099.0	1972.33-12444.19
HIP 88962	1.372 ± 0.004	0.91	-3.91	—	—	—
HIP 89320	1.586 ± 0.197	1.31	-3.65	—	—	—
HIP 89449	1.323 ± 0.019	0.82	-3.99	—	—	—
HIP 90306	1.442 ± 0.02	0.78	-3.93	—	—	—
HIP 93072	1.437 ± 0.004	0.98	-3.84	—	—	—
HIP 98007	0.978 ± 0.003	0.17	-4.99	—	—	—
HIP 99969	1.14 ± 0.015	0.62	-4.27	—	—	—
HIP 100133	1.43 ± 0.012	0.82	-3.92	—	—	—
HIP 101852	0.597 ± 0.012	0.16	-4.88	6452.83	3795.0-9658.97	1805.77-12359.28
HIP 102119	1.121 ± 0.009	1.9	-3.81	—	—	—
HIP 104687	0.654 ± 0.041	0.25	-4.92	7504.1	4539.92-10663.71	2197.54-12545.56
HIP 105504	1.255 ± 0.015	1.39	-3.81	—	—	—
HIP 107317	1.522 ± 0.021	1.73	-3.55	—	—	—

Target	Adopted B-V	S	$\log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 108036	0.378 ± 0.011	0.14	-4.84	—	—	—
HIP 109807	1.252 ± 0.005	1.21	-3.87	—	—	—
HIP 110106	1.399 ± 0.02	1.26	-3.75	—	—	—
HIP 110401	1.208 ± 0.013	0.59	-4.23	—	—	—
HIP 110663	0.77 ± 0.005	0.35	-4.91	7152.13	4286.34-10332.43	2063.3-12484.35
HIP 111976	1.548 ± 0.1	1.07	-3.74	—	—	—
HIP 113159	0.469 ± 0.005	0.08	-4.81	4922.09	2773.17-8264.65	1281.48-11983.6
HIP 115280	0.468 ± 0.004	0.13	-4.83	5405.28	3088.32-8715.42	1441.52-12121.08
HIP 115411	0.7 ± 0.005	0.09	-4.95	8010.07	4920.15-11092.03	2401.41-12683.1
HIP 116973	1.085 ± 0.008	0.55	-4.39	—	—	—
HIP 117159	0.859 ± 0.006	0.1	-5.13	—	—	—
HIP 117235	1.376 ± 0.133	0.12	-4.78	—	—	—
HIP 117410	1.244 ± 0.014	2.99	-3.49	—	—	—
HIP 117795	1.21 ± 0.015	0.55	-4.26	—	—	—
HIP 118086	1.3 ± 0.015	0.82	-4.0	—	—	—
HIP 118310	1.187 ± 0.052	0.76	-4.14	—	—	—

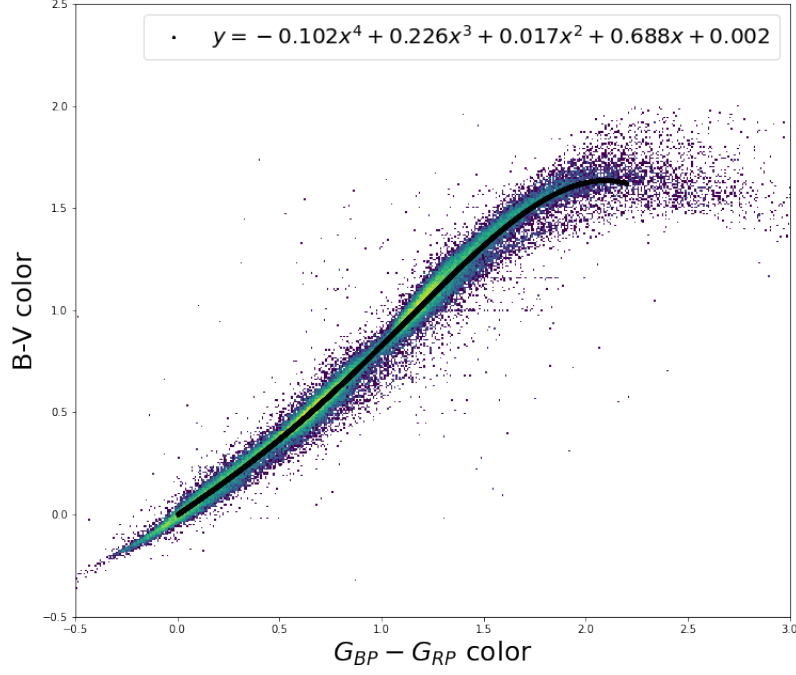


Fig. .1.— We transform *Gaia* $G_{BP} - G_{RP}$ color to Johnson $B - V$ using a polynomial fit to a large sample of *Gaia* – *Hipparcos* stars (??). We use our derived transformation when calculating median ages 68% and 95% confidence intervals from BAFFLES (?) for stars without precise $B - V$ measurements.

Target	T_{eff}	P_{rot}	Age	68% CI	95% CI
	(K)	(d)	(Myr)	(Myr)	(Myr)

Table .3:: Rotation period measurements for 23 of our 166 accelerating stars. An additional 106 stars have a TESS light curve but display either no variability signal or a signal that is not due to rotation, and 37 stars do not have a TESS light curve. Median ages and 68% and 95% confidence intervals are given for stars with a rotation period measurement that are also within the `gyro-interp` T_{eff} range.

Target	T_{eff}	P_{rot}	Age	68% CI	95% CI
	(K)	(d)	(Myr)	(Myr)	(Myr)

HIP 143	4776 ± 200	no LC	—	—	—
HIP 669	5955 ± 343	no signal	—	—	—
HIP 1068	3401 ± 51	no signal	—	—	—
HIP 1412	3807 ± 82	5.1 ± 0.44	134.6	41.14 - 385.6	6.73 - 568.83
HIP 1539	3956 ± 153	no LC	—	—	—
HIP 1771	4055 ± 154	no signal	—	—	—
HIP 2426	4104 ± 151	no signal	—	—	—
HIP 2843	6465 ± 335	no signal	—	—	—
HIP 3008	3891 ± 121	5.11 ± 0.36	128.37	39.61 - 378.36	6.54 - 563.46
HIP 3293	6442 ± 344	no signal	—	—	—
HIP 3633	4831 ± 211	no signal	—	—	—

	Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	
HIP 3724	3873 ± 104		no LC		—	—	—
HIP 4024	4692 ± 184		no signal		—	—	—
HIP 5319	6693 ± 324		no signal		—	—	—
HIP 5763	4370 ± 144		no signal		—	—	—
HIP 6776	6496 ± 324		no signal		—	—	—
HIP 6796	4590 ± 174		no signal		—	—	—
HIP 6833	5765 ± 305		no signal		—	—	—
HIP 6943	6464 ± 336		no signal		—	—	—
HIP 7287	4242 ± 123		no signal		—	—	—
HIP 8653	5755 ± 306		4.11 ± 0.34		220.38	76.93 - 423.96	11.88 - 1158.34
HIP 9067	4117 ± 150		no signal		—	—	—
HIP 9989	4279 ± 126		no signal		—	—	—
HIP 10050	6461 ± 337		no signal		—	—	—
HIP 10532	5149 ± 198		no signal		—	—	—
HIP 11090	7127 ± 426		not rotation		—	—	—
HIP 11815	3899 ± 129		2 periods		nan	nan	nan

	Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 12837	6293 ± 311		no signal		—	—
HIP 13681	4709 ± 188		no signal		—	—
HIP 13782	7615 ± 441		no signal		—	—
HIP 15211	4867 ± 211		no signal		—	—
HIP 16247	4453 ± 160		EB		—	—
HIP 16709	4104 ± 151		no signal		—	—
HIP 17118	6406 ± 343		no LC		—	—
HIP 17213	4650 ± 177		no LC		—	—
HIP 18527	5139 ± 201		9.59 ± 1.78		734.8	412.07 - 1053.55
HIP 19739	3818 ± 88		7.84 ± 1.06		254.07	129.59 - 453.05
HIP 19912	4550 ± 170		no signal		—	—
HIP 20419	4438 ± 158		10.15 ± 2.02		674.7	284.69 - 1180.52
HIP 20648	8718 ± 670		no signal		—	—
HIP 21066	6411 ± 344		not rotation		—	—
HIP 21152	6629 ± 311		no signal		—	—
HIP 22361	7155 ± 428		not rotation		—	—

	Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	
HIP 23208	4268 $\pm$ 125		no LC		—	—	—
HIP 24017	6579 $\pm$ 307		no signal		—	—	—
HIP 24177	3687 $\pm$ 87		no signal		—	—	—
HIP 24292	4342 $\pm$ 137		no signal		—	—	—
HIP 24457	4627 $\pm$ 177		no signal		—	—	—
HIP 25606	5109 $\pm$ 184		no signal		—	—	—
HIP 27021	6123 $\pm$ 315		no signal		—	—	—
HIP 28823	6887 $\pm$ 322		no signal		—	—	—
HIP 29208	5112 $\pm$ 186		no LC		—	—	—
HIP 33282	4200 $\pm$ 137		9.87 $\pm$ 1.77		507.65	220.38 - 1073.73	127.15 - 1510.58
HIP 33368	4299 $\pm$ 128		contaminated pixel		—	—	—
HIP 34804	4517 $\pm$ 166		no LC		—	—	—
HIP 36081	5957 $\pm$ 343		no signal		—	—	—
HIP 37267	4329 $\pm$ 135		no signal		—	—	—
HIP 40518	4435 $\pm$ 157		no LC		—	—	—
HIP 41274	5058 $\pm$ 173		not rotation		—	—	—

	Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	
HIP 41277	4550 $\pm$ 170		3.7 $\pm$ 0.24		95.67	30.09 - 315.98	5.16 - 532.3
HIP 41319	6511 $\pm$ 319		no signal		—	—	—
HIP 42231	4200 $\pm$ 137		no LC		—	—	—
HIP 42343	6306 $\pm$ 316		no signal		—	—	—
HIP 42507	3955 $\pm$ 153		no signal		—	—	—
HIP 43745	3936 $\pm$ 148		5.82 $\pm$ 0.56		158.14	55.2 - 396.73	8.77 - 574.25
HIP 44162	6499 $\pm$ 323		no signal		—	—	—
HIP 44953	5322 $\pm$ 239		no signal		—	—	—
HIP 45621	5150 $\pm$ 197		5.48 $\pm$ 0.51		130.82	39.99 - 287.4	6.6 - 419.96
HIP 45731	3709 $\pm$ 87		EB		—	—	—
HIP 46385	3529 $\pm$ 57		10.33 $\pm$ 1.69		—	—	—
HIP 47013	6286 $\pm$ 309		no signal		—	—	—
HIP 47261	4334 $\pm$ 136		contaminated pixel		—	—	—
HIP 47300	7661 $\pm$ 466		no signal		—	—	—
HIP 47387	4657 $\pm$ 178		no LC		—	—	—
HIP 47539	3954 $\pm$ 153		no LC		—	—	—

	Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	
HIP 47741	3363 ± 57		no signal		—	—	—
HIP 48165	4854 ± 214		no signal		—	—	—
HIP 48629	3961 ± 155		no signal		—	—	—
HIP 49046	3835 ± 91		no signal		—	—	—
HIP 49526	3713 ± 87		no signal		—	—	—
HIP 49577	4133 ± 151		EB		—	—	—
HIP 49910	3902 ± 132		no signal		—	—	—
HIP 51208	4384 ± 146		no LC		—	—	—
HIP 52339	3859 ± 96		no signal		—	—	—
HIP 53175	4211 ± 126		not rotation		—	—	—
HIP 53869	4937 ± 197		5.9 ± 0.51		102.24	32.77 - 263.89	5.56 - 436.19
HIP 54094	4839 ± 212		no signal		—	—	—
HIP 54199	5114 ± 187		no LC		—	—	—
HIP 55192	5301 ± 229		no signal		—	—	—
HIP 55409	5735 ± 307		no signal		—	—	—
HIP 57361	3516 ± 54		no signal		—	—	—

	Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 57571	3933 $\pm$ 147		no signal		—	—
HIP 57645	5093 $\pm$ 174		no signal		—	—
HIP 57984	4214 $\pm$ 123		no signal		—	—
HIP 59000	4101 $\pm$ 151		6.56 $\pm$ 0.87	185.8	82.99 - 412.07	12.82 - 574.25
HIP 59198	3831 $\pm$ 89		no LC		—	—
HIP 59199	6799 $\pm$ 318		not rotation		—	—
HIP 59247	4166 $\pm$ 153		no signal		—	—
HIP 59767	6943 $\pm$ 327		not rotation		—	—
HIP 59905	4551 $\pm$ 170		no signal		—	—
HIP 59953	3801 $\pm$ 79		no signal		—	—
HIP 60829	5636 $\pm$ 323		no signal		—	—
HIP 61947	3853 $\pm$ 96		no signal		—	—
HIP 62325	4937 $\pm$ 197		no LC		—	—
HIP 62627	4799 $\pm$ 205		no signal		—	—
HIP 63419	5278 $\pm$ 216		9.84 $\pm$ 1.5	847.11	563.46 - 1115.23	251.67 - 1373.92
HIP 63421	3894 $\pm$ 124		no signal		—	—

	Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 63661	4120 $\pm$ 150		no signal		—	—
HIP 65120	3760 $\pm$ 78		no LC		—	—
HIP 65602	5060 $\pm$ 172		no signal		—	—
HIP 65651	4355 $\pm$ 140		4.86 $\pm$ 0.4	106.19	33.08 - 337.67	5.62 - 542.49
HIP 65706	3964 $\pm$ 156		no signal		—	—
HIP 66262	4470 $\pm$ 161		no signal		—	—
HIP 66315	4580 $\pm$ 173		no signal		—	—
HIP 66587	3824 $\pm$ 89		no LC		—	—
HIP 66828	4186 $\pm$ 150		4.39 $\pm$ 0.28	107.2	33.08 - 344.13	5.62 - 547.66
HIP 69311	4399 $\pm$ 150		6.45 $\pm$ 0.62	129.59	43.96 - 357.44	7.19 - 547.66
HIP 69333	4416 $\pm$ 153		no signal		—	—
HIP 69860	3729 $\pm$ 84		no signal		—	—
HIP 69963	3762 $\pm$ 77		no signal		—	—
HIP 70472	4126 $\pm$ 150		no LC		—	—
HIP 71243	6550 $\pm$ 310		no signal		—	—
HIP 71515	4999 $\pm$ 185		no LC		—	—

	Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	
HIP 71899	6268 $\pm$ 311		2.53 $\pm$ 0.11		—	—	—
HIP 71957	6497 $\pm$ 324		no signal		—	—	—
HIP 73121	4924 $\pm$ 200		no LC		—	—	—
HIP 73183	4094 $\pm$ 152		no signal		—	—	—
HIP 73252	4045 $\pm$ 154		no signal		—	—	—
HIP 73787	4294 $\pm$ 128		no LC		—	—	—
HIP 74926	4266 $\pm$ 125		no LC		—	—	—
HIP 79203	6599 $\pm$ 305		no LC		—	—	—
HIP 79593	4260 $\pm$ 125		no LC		—	—	—
HIP 81655	3551 $\pm$ 57		no signal		—	—	—
HIP 82260	5473 $\pm$ 331		no LC		—	—	—
HIP 83676	4871 $\pm$ 211		no LC		—	—	—
HIP 83929	4177 $\pm$ 154		6.19 $\pm$ 0.63		162.7	60.69 - 392.99	9.55 - 568.83
HIP 87370	5831 $\pm$ 319		no LC		—	—	—
HIP 88962	3949 $\pm$ 151		no LC		—	—	—
HIP 89320	4206 $\pm$ 131		contaminated pixel		—	—	—

	Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 89449	4190 $\pm$ 147		no signal		—	—
HIP 90306	3767 $\pm$ 77		no signal		—	—
HIP 93072	3824 $\pm$ 89		no LC		—	—
HIP 98007	4765 $\pm$ 198		no signal		—	—
HIP 99969	4467 $\pm$ 161		no LC		—	—
HIP 100133	3769 $\pm$ 76		no LC		—	—
HIP 101852	6004 $\pm$ 335		6.87 $\pm$ 0.54		913.87	493.41 - 1914.67
HIP 102119	4217 $\pm$ 121		no LC		—	—
HIP 104687	5832 $\pm$ 319		no LC		—	—
HIP 105504	4228 $\pm$ 122		no LC		—	—
HIP 107317	3432 $\pm$ 45		6.18 $\pm$ 0.51		—	—
HIP 108036	6760 $\pm$ 321		no LC		—	—
HIP 109807	4279 $\pm$ 126		no LC		—	—
HIP 110106	3841 $\pm$ 93		no signal		—	—
HIP 110401	4240 $\pm$ 123		no signal		—	—
HIP 110663	5420 $\pm$ 289		7.3 $\pm$ 0.55		457.37	263.89 - 619.51

	Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	
HIP 111976	3904 $\pm$ 134		no signal		—	—	—
HIP 113159	6433 $\pm$ 347		no signal		—	—	—
HIP 115280	6427 $\pm$ 348		no signal		—	—	—
HIP 115411	5515 $\pm$ 331		no signal		—	—	—
HIP 116973	4614 $\pm$ 177		no signal		—	—	—
HIP 117159	5131 $\pm$ 197		no signal		—	—	—
HIP 117235	3947 $\pm$ 151		no signal		—	—	—
HIP 117410	4174 $\pm$ 154		2 periods		254.07	64.24 - 625.41	10.11 - 1400.23
HIP 117795	4342 $\pm$ 137		no signal		—	—	—
HIP 118086	4035 $\pm$ 155		no signal		—	—	—
HIP 118310	4501 $\pm$ 165		no signal		—	—	—

